Characterizing Client-Independent Stationary Bias for SCAFFOLD and FedAvg in Federated Learning

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Abstract

Even when all clients optimize the same strongly convex objective, the global model parameters produced by constant-step federated learning may still have a stationary mean that differs from the minimizer. Does increasing the number of clients eliminate this bias, or can some components persist? Answering this question clarifies the limits of client averaging. To this end, we study two federated learning algorithms, SCAFFOLD and matched FedAvg, under full participation, a sufficiently smooth objective, and independent bounded additive gradient noise. Analyzing global stationary balance and within-round local fluctuations yields multidimensional expansions with a shared γ/N term and explicit client-independent γ^2 coefficients, where γ is the step size and N is the client count. The SCAFFOLD coefficient combines the ordinary local-update contribution found in FedAvg with an additional persistent-control contribution. Although zero-sum controls cancel from the direct server increment, they change local second moments, which interact with third-order curvature to shift the stationary mean. For fixed dimension d and local-step count H, a remainder of $O_{H,d}(\gamma^2/N+\gamma^3)$, uniform in N for sufficiently small γ, identifies the second-order coefficients along arbitrary joint small-step, large-client limits after removing the shared leading term. Controlled experiments assess the predicted coefficients. In real-data analyses, finite-tail mean displacement can account for a substantial share of averaged-model parameter error, and a post-hoc comparison shows directional agreement with the derived curvature–noise structure.

1 Introduction

Federated learning enables multiple clients to collaboratively train a shared model while keeping their data distributed. To reduce communication costs, clients may perform several local stochastic-gradient updates before synchronizing their models with a central server. Although clients typically solve the same learning task using a common loss function, differences in their local data distributions can lead to different local optimization objectives and client drift. SCAFFOLD, introduced by Karimireddy et al. [6], uses control variates to mitigate this drift.

Even when all clients optimize the same objective function, constant-step stochastic updates can produce a systematic parameter displacement. The global model continues to fluctuate, and its expected parameters under the stationary distribution may differ from the objective minimizer. This difference is the *stationary mean bias*. Increasing the number of clients reduces the variance of independent gradient noise averaged at the server, but local updates also change where gradients are evaluated before aggregation. For nonlinear objectives, these local fluctuations can affect both the mean and the variance [4]. Therefore, understanding stationary mean bias requires separating contributions that diminish as the number of clients increases from the client-independent effects of local stochastic updates and persistent controls. This distinction also matters for temporal model averaging, which combines successive global models across rounds.

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Under stationarity, temporal averaging preserves the expected parameter while potentially reducing variance; thus, a small squared-bias share of single-checkpoint error need not remain small for an averaged model.

Recent stationary analyses provide a starting point. In the homogeneous setting, Mangold et al. [9, Theorem 5.3] identify a leading SCAFFOLD mean-bias contribution proportional to γ/N, where γ is the step size and N the number of clients, and point to higher-order bias not suppressed by client averaging. This expansion and the FedAvg result of Mangold et al. [10, Theorem 3] leave the client-independent γ^2 coefficient within the remainder. Consequently, these results do not identify the algorithm-dependent difference at this order or separate the contribution of ordinary local updates from that of persistent controls. We further compare their precise statements with ours in Section 2.

This study compares SCAFFOLD and FedAvg under a common strongly convex and sufficiently smooth objective, full participation, and independent bounded additive gradient noise. Matching the objective, noise laws, step size, client count, and number of local updates isolates their algorithmic difference. Our analysis combines global stationary balance with the second moments of within-round local trajectories to identify the persistent-control contribution.

Objective homogeneity also separates stochastic control effects from the correction of client drift. The controls can remain random under stochastic gradients even when all clients share the same objective, so this setting retains the mechanism of interest. Proposition 4.4 further extends the SCAFFOLD characterization to clients whose objectives differ by fixed linear terms: translating the controls absorbs these gradient offsets while preserving the global stationary law. In this class, correcting objective-induced drift and retaining stochastic mean bias coexist.

The two algorithms share the same γ/N contribution but have different formulas for the client-independent γ^2 coefficient. Let $B_{20}^{\mathcal{A}}(H)$ denote this coefficient for algorithm $\mathcal{A}$, where H is the number of local updates per communication round. The subscript 20 records the respective powers of γ and $1/N$. We obtain the decomposition

$$B_{20}^{\mathrm{SCAFFOLD}}(H) = B_{20}^{\mathrm{FedAvg}}(H) + B_{20}^{\mathrm{ctrl}}(H), \tag{1}$$

where the FedAvg coefficient is the ordinary local-update contribution and $B_{20}^{\mathrm{ctrl}}(H)$ is the additional persistent-control contribution.

Equation (1) distinguishes mechanisms that the shared leading term does not resolve. Zero-sum controls cancel from the direct server increment but alter local second moments, whose interaction with third-order curvature produces an additional mean shift. The formulas quantify its curvature–noise geometry and local-step dependence, and the cancellation at aggregation does not imply identical stationary means.

For fixed dimension d and local-step count H, we bound the remainder by $O_{H,d}(\gamma^2/N + \gamma^3)$, uniformly in N for sufficiently small γ. Subtracting the shared γ/N term and dividing by γ^2 leaves an error $O_{H,d}(1/N + \gamma)$. It vanishes along any joint limit $\gamma \to 0$, $N \to \infty$, without a relative-rate condition, separating the coefficient from the unresolved remainder.

Our contributions are:

- **Explicit stationary-bias expansions.** We derive multidimensional expansions for SCAFFOLD and matched FedAvg, with explicit γ/N and client-independent γ^2 coefficients and an N-uniform remainder, allowing noncommuting Hessian and noise covariance.
- **Mechanism attribution and an exact extension.** We separate local-update and persistent-control contributions through matched coefficients and local second moments, and extend the SCAFFOLD result to affine objective shifts.
- **Controlled tests and real-data interpretation.** We assess coefficient predictions, contraction cancellation, and covariance-matched noise laws in controlled experiments, and examine minibatch allocation, model averaging, and post-hoc directional agreement in real-data finite-tail studies.

Significance and scope. The shared leading γ/N term cannot distinguish SCAFFOLD from matched FedAvg, while existing stationary expansions leave their client-independent γ^2 coefficients unresolved. Our explicit coefficients and N-uniform remainder make this algorithm-dependent difference identifiable in joint small-step, large-client limits. The resulting decomposition quantifies the additional H-dependent contribution of persistent controls beyond ordinary local-update bias. By isolating this effect under homogeneous objectives and additive noise, the analysis provides a reference for examining additional interactions in more general federated settings.

2 Related work

Federated optimization and stationary-bias characterization. SCAFFOLD uses control variates to mitigate heterogeneity-induced client drift [6]. Finite-time analyses study LocalSGD and SCAFFOLD under gradient and Hessian similarity, weak convexity, and higher-order smoothness [7], and establish sharper SCAFFOLD convergence guarantees on quadratic objectives [8].

Stochastic local updates can induce bias even with identical client objectives. Glasgow et al. [4] study *iterate bias*, which is the deviation of the expected SGD trajectory from deterministic gradient descent with the same initialization. Their finite-step bounds and continuous perspective connect this bias to nonlinearity and noise under higher-order smoothness assumptions. Our estimand is instead the invariant-law global mean relative to the minimizer.

Stationary distributions provide a complementary perspective on constant-step stochastic optimization [2, 3]. For federated learning, Mangold et al. [10] derive stationary bias and variance expansions for FedAvg, while Mangold et al. [9] analyze the joint global-model/control process of SCAFFOLD. Table 1 compares the analytical resolution of these results with ours for the same algorithms and stationary-bias quantities.

The comparison specializes the prior results to a common objective and bounded additive noise, with the shared coefficient B_{11} given in Equation (16). The FedAvg Theorem 3 row additionally assumes identical client noise laws (Assumption A6); its Theorem 4, specialized to $f_c = f$, allows distinct noise laws and has no heterogeneity contribution.

Table 1: Analytical resolution beyond the shared leading bias term. Each entry characterizes $b^{\mathcal{A}}_{\gamma,N,H} - (\gamma/N)B_{11}$ in the common-objective, additive-noise setting. Prior bounds leave the client-independent second-order coefficient unresolved; the last row identifies it explicitly and bounds the residual.

Analysis	Characterization beyond the shared leading term
SCAFFOLD, Mangold et al. [9, Theorem 5.3]	$O(\gamma^2 H + \gamma^{3/2})$
FedAvg, Mangold et al. [10, Theorem 3]	$O(\gamma^2 H + \gamma^{3/2})$
FedAvg, Mangold et al. [10, Theorem 4], with $f_c = f$	$O(\gamma^2 H^2 + \gamma^{3/2} H)$
This work, Theorems 4.1 and 4.2	$\gamma^2 B^{\mathcal{A}}_{20}(H) \; + O_{H,d}(\gamma^2/N + \gamma^3)$

Dividing the prior remainder bounds by γ^2 gives $O(H + \gamma^{-1/2})$ or $O(H^2 + H\gamma^{-1/2})$, so neither identifies the second-order coefficient. Under the additional assumptions of Section 3, our normalized remainder is $O_{H,d}(1/N + \gamma)$, uniformly in N for fixed H, d, and vanishes in every joint limit $\gamma \to 0$, $N \to \infty$. The explicit coefficients thus identify the direction, magnitude, and local-step dependence of the client-independent contribution, separating ordinary local updates from the additional persistent-control effect.

Mangold et al. [9, Lemma 5.1] already identify the leading stationary control covariance, $(1 - 1/N)C/H + O_H(\gamma)$ in the common-objective, common-covariance setting. Our analysis propagates

this control-fluctuation term through local second moments into the matched stationary mean coefficients, with a common N-uniform remainder.

Mechanisms and attribution of stationary bias. Related work has examined distinct sources of bias: Allmeier and Gast [1] derive a Lyapunov-based first-order expression for time-averaged bias under Markovian noise; Huo et al. [5] separate memory, nonlinearity, and their interaction in nonlinear stochastic approximation; and Versini et al. [11] distinguish stochasticity and decentralization in decentralized SGD. Our analysis instead quantifies the additional stationary mean-bias coefficient attributable to persistent controls through a matched SCAFFOLD–FedAvg comparison.

3 Setting and preliminaries

We study SCAFFOLD and FedAvg in a matched, full-participation setting with identical client objectives and bounded additive gradient noise. Both algorithms use the same client noise laws and local-update configuration; the key difference is that SCAFFOLD maintains persistent client control vectors.

3.1 Objective and stochastic-gradient model

Let $x \in \mathbb{R}^d$ denote the global model parameter vector. All $N \geq 2$ clients optimize the same objective $f : \mathbb{R}^d \to \mathbb{R}$ and participate in every communication round. Each client takes $H \geq 2$ local steps of size $\gamma > 0$ per round. The dimension d and local-step count H are fixed, while N and γ may vary.

We assume $f \in C^5(\mathbb{R}^d)$, with uniformly bounded third-, fourth-, and fifth-order derivatives, and fixed constants $0 < \mu \leq L < \infty$ such that

$$\mu I_d \preceq \nabla^2 f(x) \preceq L I_d \qquad \text{for every } x \in \mathbb{R}^d. \tag{2}$$

Thus f has a unique minimizer $x^\star$. We translate it to the origin, so $x^\star = 0$ and $\nabla f(0) = 0$.

At local step h of round t, client c uses the stochastic-gradient oracle

$$\nabla F_{t,c,h}(y) = \nabla f(y) + \varepsilon_{t,c,h}, \qquad \mathbb{E}[\varepsilon_{t,c,h}] = 0, \qquad \mathbb{E}[\varepsilon_{t,c,h}\varepsilon_{t,c,h}^\top] = C \succeq 0. \tag{3}$$

For each fixed N, client c has a noise law $\nu_{N,c}$ that is unchanged across rounds and local steps and does not depend on γ. Noise vectors are mutually independent across clients, rounds, and local steps; each round's fresh noise array is independent of the complete round-start state.

The laws may differ across clients, and coordinates within a noise vector may be correlated. All clients nevertheless share the covariance C and the almost-sure bound $\|\varepsilon_{t,c,h}\| \leq K_\varepsilon$, where C and K_ε are independent of N and γ.

We define

$$A := \nabla^2 f(0) = A^\top \succ 0, \qquad T := \nabla^3 f(0). \tag{4}$$

Then, for any symmetric matrix $M \in \mathbb{R}^{d\times d}$, the vector $T[M] \in \mathbb{R}^d$ is the contraction

$$(T[M])_i := \sum_{j,k=1}^{d} T_{ijk} M_{jk}. \tag{5}$$

Finally, by letting S_C denote the unique symmetric solution of the Lyapunov equation, we have

$$AS_C + S_C A = C, \tag{6}$$

which exists because $A \succ 0$. These quantities describe the local curvature, gradient noise, and leading stationary fluctuations appearing in the main results.

3.2 SCAFFOLD updates

At the start of a round, the server holds x and client c has a control vector ξ_c. We use the zero-sum parametrization

$$\sum_{c=1}^{N} \xi_c = 0. \tag{7}$$

Here, ξ_c is the correction added to client c's stochastic gradient. It remains fixed during the H local steps and is updated only after server aggregation. Starting from $\theta_{c,0} = x$, each client updates using

$$\theta_{c,h+1} = \theta_{c,h} - \gamma(\nabla f(\theta_{c,h}) + \xi_c + \varepsilon_{c,h+1}), \qquad h = 0, \dots, H-1. \tag{8}$$

We suppress the round index when considering one round. The server averages the final local models and refreshes the controls:

$$x^+ = \frac{1}{N}\sum_{c=1}^{N} \theta_{c,H}, \qquad \xi_c^+ = \xi_c + \frac{\theta_{c,H} - x^+}{\gamma H}, \tag{9}$$

where $+$ denotes the next round-start state. Equation (9) preserves the zero-sum condition in Equation (7) exactly. This is the full-participation zero-sum representation of SCAFFOLD used by Mangold et al. [9].

3.3 Matched FedAvg updates

For comparison, the matched FedAvg algorithm uses the same objective, client noise laws, N, H, and γ, but maintains no control state. Matching concerns the stochastic setting and local-update configuration, not necessarily identical noise realizations. Starting from $\theta_{c,0}^{\text{FedAvg}} = x$, its local and server updates are

$$\theta_{c,h+1}^{\text{FedAvg}} = \theta_{c,h}^{\text{FedAvg}} - \gamma(\nabla f(\theta_{c,h}^{\text{FedAvg}}) + \varepsilon_{c,h+1}), \quad h = 0, \dots, H-1, \tag{10}$$

$$x^+ = \frac{1}{N}\sum_{c=1}^{N} \theta_{c,H}^{\text{FedAvg}}. \tag{11}$$

3.4 Stationary mean bias and coefficient notation

For sufficiently small γ, the two round-update chains admit unique invariant laws [9, 10]. Denote by $\pi_{\gamma,N,H}^{\text{SCAFFOLD}}$ the invariant law of the joint global-model/control state, and by $\pi_{\gamma,N,H}^{\text{FedAvg}}$ that of the FedAvg global model. We study the stationary mean biases of the *global model parameters*:

$$b_{\gamma,N,H}^{\text{SCAFFOLD}} := \mathbb{E}_{\pi_{\gamma,N,H}^{\text{SCAFFOLD}}}[x] - x^\star = \mathbb{E}_{\pi_{\gamma,N,H}^{\text{SCAFFOLD}}}[x], \tag{12}$$

$$b_{\gamma,N,H}^{\text{FedAvg}} := \mathbb{E}_{\pi_{\gamma,N,H}^{\text{FedAvg}}}[x] - x^\star = \mathbb{E}_{\pi_{\gamma,N,H}^{\text{FedAvg}}}[x]. \tag{13}$$

Unless stated otherwise, within-round expectations include both the stationary round-start state and the fresh noise of that round.

We index the coefficients in our expansions by powers of γ and N^{-1}. Specifically, B_{11} denotes the γ/N coefficient, while $B_{20}^{\mathcal{A}}(H)$ denotes the client-independent γ^2 coefficient for $\mathcal{A} \in \{\text{SCAFFOLD}, \text{FedAvg}\}$. Since d is fixed, we omit a dimension superscript. All results below hold for $0 < \gamma \le \gamma_0$, where $\gamma_0 > 0$ may depend on the fixed objective/noise regularity constants, H, and d, but not on N; all stated $O_{H,d}(\cdot)$ constants are uniform over $N \ge 2$ in this regime.

4 Main results

The main results resolve two contributions to stationary mean bias: a client-averaged term proportional to γ/N and a client-independent term proportional to γ^2. We first establish the SCAFFOLD expansion, then compare it with matched FedAvg to separate the effect of local stochastic updates from that of persistent controls. Throughout this section, d and $H \geq 2$ are fixed, and the assumptions of Section 3 hold. The proof details for the theorems, corollaries, and propositions in this section are provided in Appendix A and B.

4.1 Two bias scales in SCAFFOLD

Server averaging reduces fluctuations across independent clients, but nonlinear effects accumulated within local trajectories need not vanish as N increases. The following theorem makes the two resolved bias scales explicit.

Theorem 4.1 (Stationary mean-bias expansion for SCAFFOLD)**.** *Under the assumptions of Section 3 and the* SCAFFOLD *updates in Equations (8) and (9), there exist constants $\gamma_0 > 0$ and $C_{H,d}^{\mathrm{SCAFFOLD}} < \infty$, independent of N, such that for every $N \geq 2$ and $0 < \gamma \leq \gamma_0$,*

$$b_{\gamma,N,H}^{\mathrm{SCAFFOLD}} = -\frac{\gamma}{2N} A^{-1} T[S_C] - \gamma^2 \frac{(H-1)(5H-1)}{12H} A^{-1} T[C] + R_{\gamma,N,H}^{\mathrm{SCAFFOLD}}, \tag{14}$$

where

$$\|R_{\gamma,N,H}^{\mathrm{SCAFFOLD}}\| \leq C_{H,d}^{\mathrm{SCAFFOLD}} \left(\frac{\gamma^2}{N} + \gamma^3 \right). \tag{15}$$

The expansion distinguishes a term proportional to γ/N from one proportional to γ^2 with an N-independent coefficient:

$$B_{11} = -\frac{1}{2} A^{-1} T[S_C], \qquad B_{20}^{\mathrm{SCAFFOLD}}(H) = -\frac{(H-1)(5H-1)}{12H} A^{-1} T[C]. \tag{16}$$

Here, B_{11} multiplies γ/N, whereas $B_{20}^{\mathrm{SCAFFOLD}}(H)$ multiplies γ^2 without a factor $1/N$. The latter is governed by $A^{-1}T[C]$: the noise covariance C interacts with the third derivative T, and A^{-1} maps the resulting vector to a parameter displacement. In particular, $T[C] = 0$ eliminates this coefficient even when the gradient noise is nonzero.

At these resolved orders, increasing N suppresses the γ/N term but not the explicit γ^2 term; reducing γ affects them linearly and quadratically, respectively. For either algorithm $\mathcal{A} \in \{\mathrm{SCAFFOLD}, \mathrm{FedAvg}\}$ with $B_{11} \neq 0$, the ratio of their magnitudes is

$$\frac{\|\gamma^2 B_{20}^{\mathcal{A}}(H)\|}{\|(\gamma/N) B_{11}\|} = \gamma N \, \frac{\|B_{20}^{\mathcal{A}}(H)\|}{\|B_{11}\|}. \tag{17}$$

Equation (17) shows that the second-order contribution need not be smaller when N is large. The full stationary bias is the sum of both vectors and the remainder, so cancellation is possible. Their relative magnitudes also do not, by themselves, determine the squared-bias share of parameter mean-squared error.

4.2 Local-update bias without controls: matched FedAvg

The SCAFFOLD coefficient alone does not reveal which part of the γ^2 term arises from local updates common to both algorithms. Matched FedAvg provides the corresponding expansion without a persistent control state.

Theorem 4.2 (Stationary mean-bias expansion for matched FedAvg)**.** *Under the assumptions of Section 3 and the matched* FedAvg *updates in Equations (10) and (11), the step-size threshold γ_0 can be chosen jointly with Theorem 4.1. There exists $C_{H,d}^{\text{FedAvg}} < \infty$, independent of N and γ, such that for $N \geq 2$ and $0 < \gamma \leq \gamma_0$,*

$$b_{\gamma,N,H}^{\text{FedAvg}} = -\frac{\gamma}{2N} A^{-1} T[S_C] - \gamma^2 \frac{H-1}{4} A^{-1} T[C] + R_{\gamma,N,H}^{\text{FedAvg}}, \tag{18}$$

where

$$\|R_{\gamma,N,H}^{\text{FedAvg}}\| \leq C_{H,d}^{\text{FedAvg}} \left(\frac{\gamma^2}{N} + \gamma^3 \right). \tag{19}$$

In particular,

$$B_{20}^{\text{FedAvg}}(H) = -\frac{H-1}{4} A^{-1} T[C]. \tag{20}$$

The two expansions share B_{11} but have different γ^2 coefficients. Thus, when $T[C] \neq 0$, local stochastic updates produce a client-independent second-order mean shift even without controls. Subtracting the coefficients then identifies the additional contribution associated with SCAFFOLD controls.

4.3 The additional contribution of persistent controls

Define the difference between the two second-order coefficients as

$$B_{20}^{\text{ctrl}}(H) := -\frac{(H-1)(2H-1)}{12H} A^{-1} T[C]. \tag{21}$$

Corollary 4.3 (Matched SCAFFOLD–FedAvg attribution)**.** *Under the assumptions of Theorems 4.1 and 4.2, we derive*

$$\boxed{B_{20}^{\text{SCAFFOLD}}(H) = B_{20}^{\text{FedAvg}}(H) + B_{20}^{\text{ctrl}}(H).} \tag{22}$$

Consequently, the following equation holds:

$$b_{\gamma,N,H}^{\text{SCAFFOLD}} - b_{\gamma,N,H}^{\text{FedAvg}} = \gamma^2 B_{20}^{\text{ctrl}}(H) + O_{H,d}\left(\frac{\gamma^2}{N} + \gamma^3 \right). \tag{23}$$

For $H \geq 2$, the FedAvg and control coefficients are positive scalar multiples of the same vector $-A^{-1}T[C]$ and therefore point in the same direction whenever nonzero. Specifically, $B_{20}^{\text{ctrl}}(H) = \frac{2H-1}{3H} B_{20}^{\text{FedAvg}}(H)$, so at $H = 2$ the control coefficient equals half the FedAvg coefficient. Equation (21) also yields $B_{20}^{\text{ctrl}}(4) = \frac{7}{2} B_{20}^{\text{ctrl}}(2)$, a factor of 3.5 between these two fixed-H cases.

The leading coefficient B_{11} has no H dependence, whereas the second-order prefactors depend on H and the remainder constants and admissible step-size threshold may also depend on it. These are comparisons of fixed local-step counts, not a uniform growing-H regime. Section 5 explains the underlying control-state mechanism.

4.4 Identifying the second-order coefficients

Identifying a γ^2 coefficient requires a remainder that vanishes after normalization by γ^2. For SCAFFOLD, we subtract the shared leading term and define

$$\mathcal{C}_{\gamma,N,H}^{\text{SCAFFOLD}} := \frac{b_{\gamma,N,H}^{\text{SCAFFOLD}} - (\gamma/N) B_{11}}{\gamma^2}. \tag{24}$$

Theorem 4.1 implies

$$\mathcal{C}^{\mathrm{SCAFFOLD}}_{\gamma,N,H} = B^{\mathrm{SCAFFOLD}}_{20}(H) + O_{H,d}\Big(\frac{1}{N} + \gamma\Big). \tag{25}$$

Because the remainder bound is uniform in N, the normalized error vanishes along *any* joint limit $\gamma \to 0$, $N \to \infty$, without a condition relating their rates. The same holds for FedAvg and the normalized matched difference. This identifies both algorithm-specific coefficients and the additional control contribution at a common asymptotic resolution.

4.5 An exact extension to affine objective heterogeneity

The homogeneous result extends exactly to a restricted form of objective heterogeneity: clients may have different local minimizers while sharing the same curvature. The key is to absorb constant gradient offsets into the SCAFFOLD control state.

Proposition 4.4 (Control translation for affine objective shifts)**.** *Let f and the additive-noise laws satisfy Section 3, and fix $N \geq 2$, $H \geq 2$ and $0 < \gamma \leq \gamma_0$. For deterministic vectors $a_c \in \mathbb{R}^d$ with $\sum_c a_c = 0$, we set*

$$f_c(x) = f(x) + a_c^\top x, \qquad \widetilde{\xi}_c = \xi_c + a_c. \tag{26}$$

We run SCAFFOLD on the zero-sum control state space, with ∇f_c in Equation (8) and the same server/control updates in Equation (9). The transformed state $(x, \widetilde{\xi})$ follows exactly the homogeneous SCAFFOLD recursion for f. Its global invariant distribution therefore coincides with that of the homogeneous chain. In particular, the expansion and remainder in Theorem 4.1 remain valid for its global stationary mean bias.

Although individual clients may have different minimizers, their average objective remains $N^{-1}\sum_c f_c = f$. With translated initial controls and shared noise realizations, the affine and homogeneous SCAFFOLD recursions coincide pathwise, although unrelated control initializations need not yield identical transients. The state translation therefore preserves the global stationary distribution and its mean-bias expansion, even in the presence of these client-specific affine shifts.

5 Why persistent controls affect stationary bias

Corollary 4.3 identifies an additional client-independent bias coefficient in SCAFFOLD relative to matched FedAvg. The effect is indirect: zero-sum controls cancel from the server update, but they change the local trajectories before aggregation. We trace this effect through the server increment, local second moments, and stationary mean balance.

5.1 Server-level cancellation does not imply identical local trajectories

Let $\bar{\varepsilon}_{h+1} := N^{-1}\sum_c \varepsilon_{c,h+1}$. Summing the local updates in Equation (8) and averaging over clients, with the zero-sum condition in Equation (7), gives

$$x^+ - x = -\gamma \sum_{h=0}^{H-1} \frac{1}{N}\sum_{c=1}^{N} \nabla f(\theta_{c,h}) - \gamma \sum_{h=0}^{H-1} \bar{\varepsilon}_{h+1}. \tag{27}$$

The controls disappear explicitly from this expression, but they still determine the local iterates $\theta_{c,h}$ at which the gradients are evaluated. Matched FedAvg has the same form of server increment, with different local trajectories because it has no control state.

5.2 Controls create an additional local second-moment contribution

To measure the effect on local trajectories, define the stationary *raw* second moments $M_x^{\mathcal{A}} = \mathbb{E}[xx^\top]$ and $M_h^{\mathcal{A}} = N^{-1}\sum_c \mathbb{E}[\theta_{c,h}\theta_{c,h}^\top]$ for algorithm $\mathcal{A}$. Their within-round difference is

$$U_h^{\mathcal{A}} := M_h^{\mathcal{A}} - M_x^{\mathcal{A}}, \qquad \mathcal{A} \in \{\text{SCAFFOLD}, \textsc{FedAvg}\}. \tag{28}$$

For $h = 0, \ldots, H-1$, the moment estimates are

$$U_h^{\textsc{FedAvg}} = \gamma^2 hC + O_{H,d}\left(\frac{\gamma^2}{N} + \gamma^3\right), \tag{29}$$

$$U_h^{\text{SCAFFOLD}} = \gamma^2\left(h + \frac{h^2}{H}\right)C + O_{H,d}\left(\frac{\gamma^2}{N} + \gamma^3\right). \tag{30}$$

The shared hC term reflects fresh gradient noise accumulated over local steps. In SCAFFOLD, the same round-start control acts throughout those steps, adding the h^2C/H term. Its quadratic dependence on h reflects this persistence; the $1/H$ factor comes from the leading stationary control covariance established by Mangold et al. [9, Lemma 5.1].

5.3 Local fluctuations are converted into a mean shift by curvature

Taking expectations of the server increment under stationarity yields the balance equation

$$0 = \sum_{h=0}^{H-1}\frac{1}{N}\sum_{c=1}^{N}\mathbb{E}[\nabla f(\theta_{c,h})]. \tag{31}$$

Because f is nonlinear, these expected gradients depend not only on the mean parameter but also on local fluctuations. A Taylor expansion around the minimizer makes the role of the second moments explicit:

$$b_{\gamma,N,H}^{\mathcal{A}} = -\frac{1}{2}A^{-1}T[M_x^{\mathcal{A}}] - \frac{1}{2H}A^{-1}T\left[\sum_{h=0}^{H-1}U_h^{\mathcal{A}}\right] + O_{H,d}\left(\frac{\gamma^2}{N} + \gamma^3\right). \tag{32}$$

Both algorithms have the same leading global second moment, $M_x^{\mathcal{A}} = (\gamma/N)S_C + O_{H,d}(\gamma^2/N + \gamma^3)$. Their additional client-independent difference therefore comes from the local moments in Equations (29) and (30). Inserting the extra γ^2h^2C/H term into Equation (32) gives

$$-\frac{\gamma^2}{2H}A^{-1}T\left[\sum_{h=0}^{H-1}\frac{h^2}{H}C\right] = \gamma^2 B_{20}^{\text{ctrl}}(H),$$

which recovers the additional coefficient in Corollary 4.3.

Thus the controls affect the stationary mean through their influence on local fluctuations, even though their direct contribution cancels at the server. The nonlinear curvature converts the difference in local second moments into the matched bias difference.

6 Numerical study

We examine the coefficient formulas, their underlying mechanism, and their implications for averaged models. Controlled additive-noise experiments isolate the individual coefficients, the extra control contribution, and its dependence on noise geometry. Shared-objective logistic-regression experiments examine how client allocation and temporal averaging affect finite-tail mean displacement. A final post-hoc comparison connects the observed displacement direction to the analytical curvature–noise vector.

6.1 Evaluation quantities and experimental settings

The controlled experiments use nonlinear strongly convex objectives with known minimizers and coefficient targets. We estimate the global mean with the stationary mean-balance identity. Subtracting the shared leading term and rescaling exposes the second-order coefficient:

$$\widehat{\mathcal{C}}_j^{\mathcal{A}} := \frac{\widehat{b}_j^{\mathcal{A}} - (\gamma/N)B_{11,j}}{\gamma^2}, \qquad j = 1, 2. \tag{33}$$

Theorems 4.1 and 4.2 predict convergence to $B_{20}^{\mathcal{A}}(H)$ as the step size decreases and the client count increases. The fitted intercept of a linear extrapolation estimates this limiting coefficient. More details regarding the experimental settings are provided in appendices. Table 7 collects the controlled parameter grids; Tables 8 and 9 give run lengths and interval procedures. The real-data settings are collected in Table 20.

6.2 Controlled tests of analytical predictions

We first estimate each algorithm's coefficient and then directly compare the algorithms. Complementary tests separate client-count dependence, multidimensional consistency, local-step dependence, and affine-shift equivalence. The final controlled experiment tests two specific predictions: cancellation when $T[C] = 0$ and invariance of the matched coefficient under covariance-preserving changes to the noise law.

6.2.1 Individual stationary-bias coefficients

Design and rationale. This experiment checks the explicit $B_{20}^{\mathrm{SCAFFOLD}}(H)$ and $B_{20}^{\mathrm{FedAvg}}(H)$ formulas separately. A two-dimensional objective with noncommuting Hessian and noise covariance provides a concrete test of the matrix-valued geometry. Along the joint path $\gamma = 1/N$, both normalized remainder scales decrease together. A smaller-step grid probes the stability of the $H = 4$ estimates.

Results and interpretation. Table 2 compares the fitted vectors with the analytical targets. For $H = 2$, both algorithms' targets fall within their simultaneous intervals. For $H = 4$, the FedAvg intervals contain the targets on both grids; the SCAFFOLD estimates move closer on the smaller-step grid, whose intervals also contain both targets. These comparisons support the predicted coefficient values while showing greater grid sensitivity for SCAFFOLD at $H = 4$. Complete interval endpoints are in Table 11. The matched-pair rows are interpreted next.

6.2.2 Matched contribution of persistent controls

Design and rationale. To isolate the additional term in Corollary 4.3, we compare SCAFFOLD and FedAvg using the same initial global model and indexed noise within each pair. Their shared leading coefficient cancels in

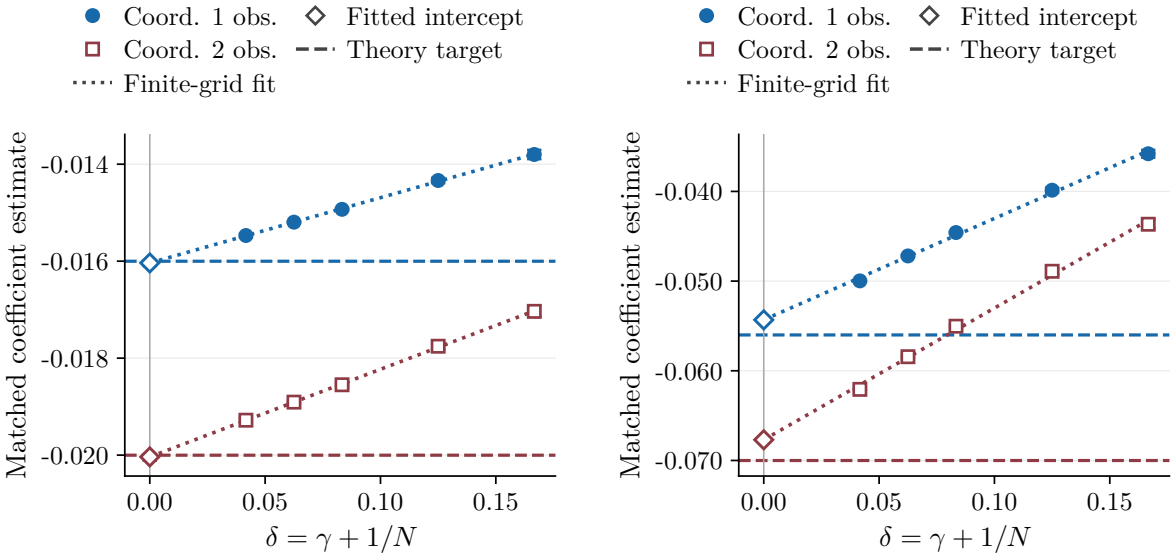
$$\widehat{D}_j := \widehat{\mathcal{C}}_j^{\mathrm{SCAFFOLD}} - \widehat{\mathcal{C}}_j^{\mathrm{FedAvg}} = \frac{\widehat{b}_j^{\mathrm{SCAFFOLD}} - \widehat{b}_j^{\mathrm{FedAvg}}}{\gamma^2}. \tag{34}$$

The target is $B_{20}^{\mathrm{ctrl}}(H)$. Common noise reduces sampling variation in the comparison, making the extra contribution easier to resolve than by subtracting independently estimated algorithm means.

Table 2: Analytical coefficient vectors and fitted finite-grid intercepts. Individual rows target $B_{20}^{\mathcal{A}}(H)$; matched-pair rows target $B_{20}^{\mathrm{ctrl}}(H)$. P and S denote the primary and smaller-step joint grids in Table 7. Values are rounded to seven decimal places; simultaneous intervals are in Table 11.

Study	H	Grid	Target vector	Estimated intercept vector
SCAFFOLD	2	P	$(-0.0480040, -0.0600050)$	$(-0.0476713, -0.0595811)$
SCAFFOLD	4	P	$(-0.1520127, -0.1900159)$	$(-0.1502614, -0.1874524)$
SCAFFOLD	4	S	$(-0.1520127, -0.1900159)$	$(-0.1514772, -0.1893273)$
FedAvg	2	P	$(-0.0320027, -0.0400033)$	$(-0.0320430, -0.0400539)$
FedAvg	4	P	$(-0.0960080, -0.1200100)$	$(-0.0959717, -0.1198059)$
FedAvg	4	S	$(-0.0960080, -0.1200100)$	$(-0.0959727, -0.1199590)$
Paired difference	2	P	$(-0.0160013, -0.0200017)$	$(-0.0160363, -0.0200366)$
Paired difference	4	P	$(-0.0560047, -0.0700059)$	$(-0.0543238, -0.0676943)$
Paired difference	4	S	$(-0.0560047, -0.0700059)$	$(-0.0559262, -0.0698957)$

Results and interpretation. Figure 1 displays the matched estimates and analytical targets. The $H = 2$ intercept intervals contain both targets. For $H = 4$, the smaller-step grid reduces absolute intercept errors by about 95%, although its intervals still exclude the targets. The experiment therefore resolves a nonzero control-associated contribution and shows improved coefficient agreement on the finer grid, with a remaining $H = 4$ offset. Appendix C.2.2 examines this offset: the grid-point residuals scale approximately with $\gamma + 1/N$, and their finite-step, finite-client, and finite-run sources remain unresolved.



(a) Two local steps

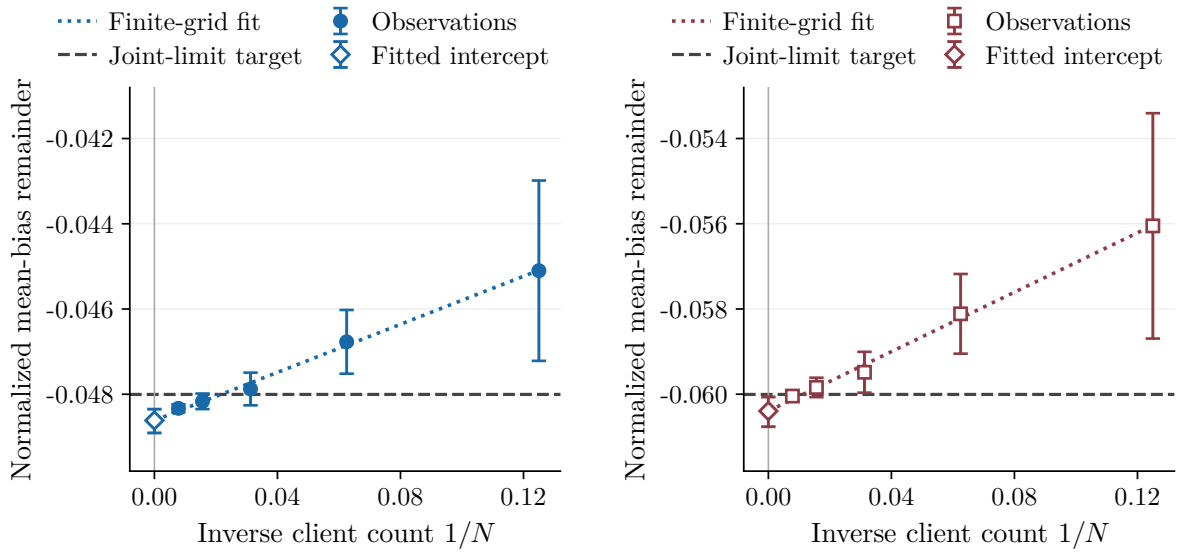
(b) Four local steps

Figure 1: Matched SCAFFOLD–FedAvg coefficient estimates for (a) $H = 2$ and (b) $H = 4$ on the primary joint grids. The ordinate is $\widehat{D}_j$ from Equation (34). Circles and squares show the two coordinates, dotted lines show finite-grid linear fits, and dashed horizontal lines give the analytical targets. Open diamonds at $\delta = 0$ mark the fitted intercepts. Pointwise bars use simultaneous Student-t intervals; intercept bars use simultaneous seed-block-bootstrap intervals.

6.2.3 Client-count dependence at fixed step size

Design and rationale. The joint-grid experiment changes N and γ together. Here, we hold the step size and local-step count fixed and increase N, isolating the effect of additional client averaging. A client-independent second-order term should remain visible as finite-client corrections to the normalized statistic decrease.

Results and interpretation. The normalized SCAFFOLD contribution remains visibly nonzero across the tested client counts (Figure 2). The fitted offsets from the joint-limit coefficient are reported in Table 13; the fixed-step expansion permits an $O(\gamma)$ offset. This result illustrates the distinction between increasing the client count and reducing the step size: additional clients suppress the client-dependent term while leaving a residual mean shift.



(a) Coordinate 1

(b) Coordinate 2

Figure 2: Client-count dependence at fixed step size, shown by coordinate. The ordinate is $\widehat{\mathcal{C}}_j^{\mathrm{SCAFFOLD}}$ from Equation (33): the mean-bias estimate after subtracting the shared γ/N term and dividing by γ^2. Dotted lines give finite-grid fits and open diamonds their intercepts at $1/N = 0$. Dashed horizontal lines give the joint-limit references $B_{20}^{\mathrm{SCAFFOLD}}(2)$; a fixed step size permits an $O(\gamma)$ offset. Bars show simultaneous Student-t intervals at grid points and seed-block-bootstrap intervals for fitted intercepts.

6.2.4 Multidimensional and coordinate consistency

Design and rationale. A five-dimensional SCAFFOLD instance tests the coefficient formula beyond the primary two-dimensional example, again using noncommuting Hessian and covariance matrices. A separate rotation check transforms the objective, noise, and initial state together. The resulting trajectories and coefficients should transform by the same orthogonal map.

Results and interpretation. All ten coordinate targets across $H = 2$ and $H = 4$ lie within the reported simultaneous intervals (Table 14). The rotation check has pathwise residuals below 1.6×10^{-14} (Appendix C.3.2). The five-dimensional results support the coefficient formula across coordinates; the rotation check confirms that the implementation respects its coordinate transformation.

Table 3: Control-coefficient estimates on the independent Cartesian grid. Each row compares the fitted intercept with $B_{20}^{\mathrm{ctrl}}(H)$. Intervals are nominal 95% run-seed intervals.

H	j	Estimate	Target	Nominal interval
2	1	−0.01599546	−0.01600134	[−0.01601159, −0.01597933]
2	2	−0.01999408	−0.02000167	[−0.02001422, −0.01997394]
4	1	−0.05595226	−0.05600469	[−0.05601109, −0.05589342]
4	2	−0.06993656	−0.07000586	[−0.07000999, −0.06986314]
8	1	−0.13962104	−0.14001171	[−0.13983973, −0.13940235]
8	2	−0.17449119	−0.17501464	[−0.17476345, −0.17421894]

6.2.5 Local-step dependence on an independent grid

Design and rationale. The theory assigns different roles to local training length: B_{11} is shared across H, whereas $B_{20}^{\mathrm{ctrl}}(H)$ changes explicitly with H. An independent implementation tests both predictions. Individual fits of $(N/\gamma)\widehat{b}$ estimate B_{11} freely. Matched fits on a Cartesian grid vary γ and $1/N$ independently, allowing a direct comparison with the predicted relation $B_{20}^{\mathrm{ctrl}}(4) = 3.5B_{20}^{\mathrm{ctrl}}(2)$.

Results and interpretation. The leading-coefficient intervals contain their targets for $H = 1, 2, 4$; both algorithms have offsets at $H = 8$. The adjusted SCAFFOLD $H = 8$ versus $H = 2$ leading-intercept contrasts exclude zero (Table 15). For the matched coefficient, Table 3 shows target inclusion at $H = 2, 4$ and target exclusion at $H = 8$. The adjusted $H = 4$ minus $3.5(H = 2)$ contrast contains zero in both coordinates. These independent estimates support the predicted local-step relation at $H = 2, 4$ and identify $H = 8$ as more sensitive to the fitted grid. Restricted-grid estimates and the separate joint-grid $H = 4$ discrepancy are documented in Appendix C.2.

6.2.6 Affine-shift equivalence

Design and rationale. Proposition 4.4 predicts exact equivalence after client-specific gradient offsets are absorbed into the controls. We test this identity by pairing affine-shift and homogeneous recursions with translated initial controls and identical noise, across several shift sizes, client counts, and local-step counts.

Results and interpretation. Across 24 configurations of 2,000 rounds, the maximum control-state residual is 1.64313×10^{-14}. The numerical recursions therefore reproduce the predicted state translation to floating-point accuracy. This illustrates how correction of affine gradient offsets can coexist with the same stochastic stationary-bias mechanism.

6.2.7 Contraction cancellation and covariance-matched noise

Design and rationale. These tests isolate the specific role of $T[C]$. A nonquadratic trigonometric objective has $T \neq 0$, and the covariance family $C_\rho = \mathrm{diag}(1 + \rho, 1 - \rho)$ preserves total noise variance. At $\rho = 0$, the contraction vanishes while $B_{11} = (-0.0125, 0)^\top$ remains nonzero, isolating cancellation of the client-independent coefficient. The positive and negative covariance controls produce opposite-signed targets. At one fixed covariance, bounded symmetric and asymmetric noise laws change third and fourth moments while preserving the predicted coefficients. Unconstrained Cartesian-grid fits estimate the individual and matched quantities; the model and laws are given in Appendix C.1.4.

Results and interpretation. Table 4 separates the matched null from both nonzero covariance controls and recovers their predicted signs and magnitudes. The noise-law contrast is small. In both coordinates, the null and law-contrast intervals fall within the prespecified coefficient-resolution margin ± 0.004375, equal to 20% of the nonzero matched target magnitude. Thus the matched increment responds to covariance geometry and agrees across covariance-matched noise laws at the reported resolution. Individual first-coordinate null and law contrasts remain imprecise; one small paired second-coordinate interval excludes zero while lying within the resolution margin. Appendix C.2.5 gives these complementary outcomes and their sensitivity analyses.

Table 4: Contraction and noise-law checks for the first coordinate of the matched coefficient at $H = 4$. The covariance controls predict positive, zero, and negative coefficients; the last row compares noise laws with identical covariance. Intervals are approximate simultaneous 95% sampling intervals for the prespecified 36-scalar family.

ρ	Noise law	Target	Estimate	Simultaneous interval
$-1/4$	Symmetric	0.021875	0.02187858	[0.02184128, 0.02191589]
0	Symmetric	0.000000	−0.00002056	[−0.00006150, 0.00002039]
$+1/4$	Symmetric	−0.021875	−0.02185115	[−0.02189536, −0.02180693]
$+1/4$	Asymmetric	−0.021875	−0.02188179	[−0.02193153, −0.02183205]
Symmetric minus asymmetric ($\rho = 1/4$)		0.000000	0.00003064	[−0.00003564, 0.00009692]

6.3 Real-data implications of a fixed sampling budget

Design and rationale. We examine whether equal aggregate noise makes client allocations interchangeable. All clients optimize the same a9a logistic-regression objective, and the total minibatch budget Nm is fixed. At the reference minimizer x_{ref}, the server-averaged noise covariance is $\Sigma_{\mathrm{ref}}/(Nm)$, where Σ_{ref} is the single-example gradient-noise covariance. The local covariance is instead Σ_{ref}/m, whose substitution into Equation (21) suggests a γ^2/m dependence. This motivates comparing allocations that have equal reference aggregate noise and different local noise levels.

We measure finite-tail displacement along the preselected unit direction $v \propto -A_{\mathrm{ref}}^{-1} T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}]$, with $A_{\mathrm{ref}} = \nabla^2 f(x_{\mathrm{ref}})$ and $T_{\mathrm{ref}} = \nabla^3 f(x_{\mathrm{ref}})$:

$$\widehat{\delta}_v^{\mathcal{A}} = v^\top (\overline{x}^{\mathcal{A}} - x_{\mathrm{ref}}), \tag{35}$$

Here $\overline{x}^{\mathcal{A}}$ averages the per-run tail means across independent runs. The statistic measures the signed mean displacement along a direction fixed before the training runs.

Results and interpretation. Displacement increases as the same sample budget is distributed among more clients with smaller local batches (Figure 3). The $N = 128$ minus $N = 8$ contrasts are 0.003114 for SCAFFOLD and 0.002072 for FedAvg, with adjusted intervals $[0.002749, 0.003478]$ and $[0.001713, 0.002431]$. Raw variance estimates remain near 0.00235–0.00236. Equal reference aggregate covariance therefore coexists with different finite-tail mean displacements: the local allocation carries information that server-level covariance alone does not capture.

The training-configuration comparison in Table 5 provides a complementary interpretation. Reducing γ lowers both displacement and raw variance; reducing H lowers displacement with much less variance change. These are comparisons under the run schedules in Table 8. The disjoint IID and label-skew results in Appendix C.4.1 additionally show how heterogeneity-induced drift correction can dominate the algorithm difference when the common-objective condition is relaxed.

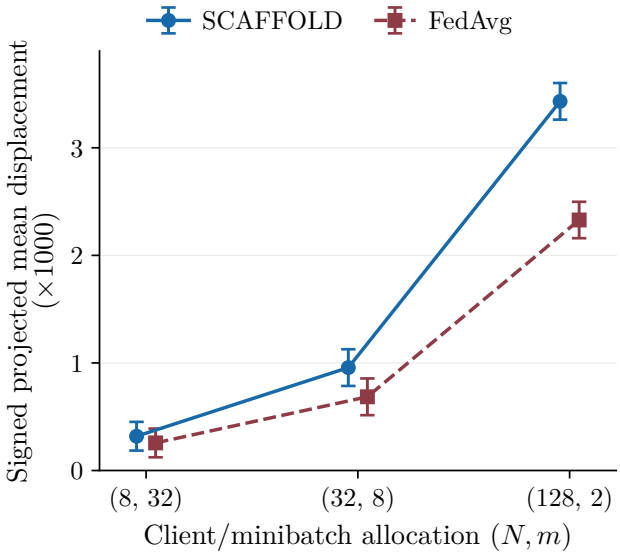


Figure 3: Finite-tail mean displacement under a fixed sampling budget. Each category gives a client-count/minibatch allocation (N, m) with $Nm = 256$. The ordinate is the signed projection in Equation (35), scaled by 10^3. Bars are nominal 95% run-seed intervals; small offsets distinguish the algorithms at the same allocation.

Table 5: Step-size and local-step comparisons for the shared objective at $N = 32, m = 8$. All entries are multiplied by 10^3. $\widehat{\delta}_v$ is signed finite-tail mean displacement and $\widehat{V}$ is raw-checkpoint parameter variance. S: SCAFFOLD; F: FedAvg.

Setting	$\widehat{\delta}_v^{\mathrm{S}}$	$\widehat{\delta}_v^{\mathrm{F}}$	$\widehat{V}^{\mathrm{S}}$	$\widehat{V}^{\mathrm{F}}$
$H = 4,\ \gamma = \gamma_b$	0.9567	0.6851	2.3621	2.3613
$H = 4,\ \gamma = \gamma_b/2$	0.3547	0.2796	1.1723	1.1722
$H = 4,\ \gamma = \gamma_b/4$	0.0622	0.0420	0.5860	0.5860
$H = 2,\ \gamma = \gamma_b$	0.3402	0.2474	2.3601	2.3599

6.4 Model averaging changes the error decomposition

Design and rationale. Under stationarity, temporal averaging preserves the expected model while potentially reducing variance. We therefore examine whether mean displacement becomes a larger share of parameter error when successive global models are averaged. We use the same shared-objective runs and average successive global models within each run over nested windows. Let $Y_{r,K}^{\mathcal{A}}$ be run r's output averaged over its last K equal-length batches. Independent runs supply repeated outputs, while the windows share a common endpoint. Table 20 gives the window lengths and computational budget.

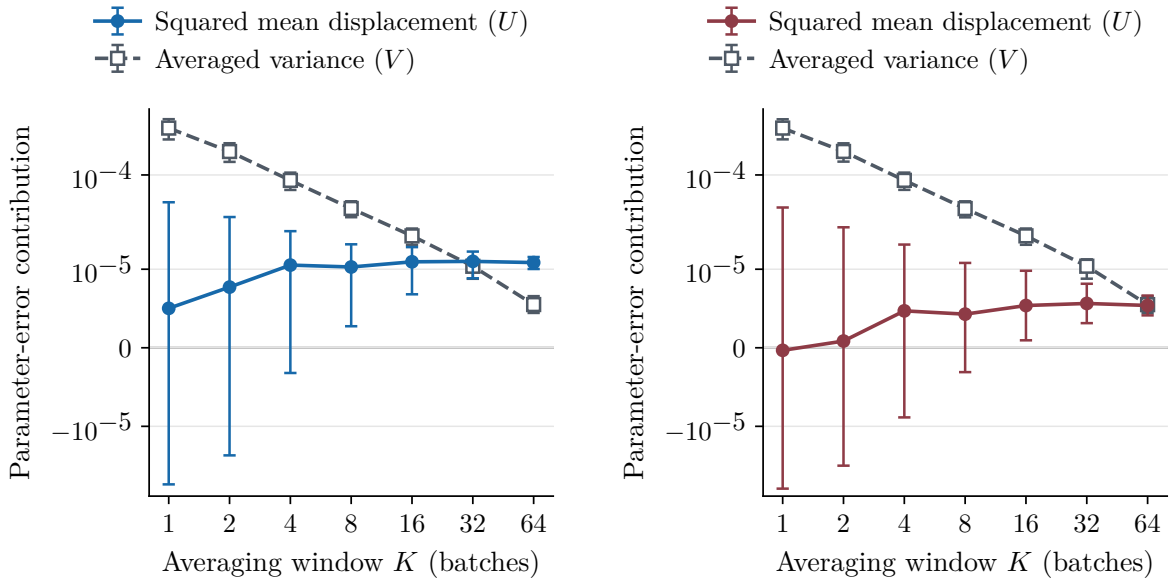
For a random output $Y_K^{\mathcal{A}}$ of this finite-window procedure,

$$\mathbb{E}\|Y_K^{\mathcal{A}} - x_{\mathrm{ref}}\|^2 = \|\mathbb{E}[Y_K^{\mathcal{A}}] - x_{\mathrm{ref}}\|^2 + \operatorname{tr}\operatorname{Cov}(Y_K^{\mathcal{A}}). \tag{36}$$

We estimate squared mean displacement by the sampling-corrected cross-run statistic $\widehat{U}_K$ and averaged-output variance by $\widehat{V}_K$. Their sum is the estimated parameter MSE. Sampling correction allows $\widehat{U}_K$ to be negative with finitely many runs; the plots retain these values. Appendix C.4.3 gives the estimators.

Results and interpretation. In the primary allocation, mean displacement accounts for only about 0.50% and 0.23% of raw-checkpoint parameter MSE for SCAFFOLD and FedAvg,

respectively. Figure 4 shows how this composition changes after averaging. From $K = 1$ to 64, averaged-output variance decreases about 57-fold. At $K = 64$, the mean-displacement shares reach 68.1% and 49.5%, with simultaneous intervals $[62.6\%, 73.7\%]$ and $[41.1\%, 57.9\%]$. Systematic displacement therefore becomes a substantial part of the averaged-model error in this experiment, even though individual checkpoints are variance-dominated.



(a) SCAFFOLD

(b) FedAvg

Figure 4: Parameter-error decomposition after temporal model averaging. The panels compare estimated squared mean displacement $\widehat{U}_K$ with averaged-output variance $\widehat{V}_K$ over windows of K batches in the primary shared-objective configuration. Each batch contains 1,280 rounds. Bars are approximate simultaneous 95% primary-family intervals. The symmetric-log axis is linear between -10^{-5} and 10^{-5} and retains negative sampling-corrected estimates and intervals crossing zero.

At the longest window, the SCAFFOLD-minus-FedAvg parameter-MSE contrast is 6.39×10^{-6}, with interval $[5.83, 6.95] \times 10^{-6}$, mainly associated with squared mean displacement. The corresponding training-gap contrast is 3.64×10^{-7}, while SCAFFOLD has lower held-out log loss by 3.27×10^{-5}. Thus parameter error and predictive loss respond differently. The averaging result concerns finite-tail outputs; Appendix C.4.3 reports the remaining temporal uncertainty.

6.5 Directional agreement beyond the additive-noise model

Design and rationale. We examine whether the curvature–noise geometry helps explain the algorithmic mean displacement under natural minibatch noise. The post-hoc comparison substitutes the reference-objective quantities from Section 6.3 into the analytical expression:

$$p_{\gamma,H,m} = -\frac{\gamma^2}{m}\,\kappa(H)\,A_{\mathrm{ref}}^{-1}T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}], \qquad \kappa(H) = \frac{(H-1)(2H-1)}{12H}. \tag{37}$$

The observed difference is $\widehat{\Delta}_K = R^{-1}\sum_r (Y_{r,K}^{\mathrm{SCAFFOLD}} - Y_{r,K}^{\mathrm{FedAvg}})$ over $R = 12$ paired runs. The proxy is computed directly from the objective and reference covariance. Comparing all 124 coordinates tests its geometric alignment with the observed mean difference. Cosine measures directional agreement; projection ratio and relative residual measure amplitude agreement and vector mismatch. Because both algorithms' second-order terms share the same analytical

direction, the amplitude and local-step comparisons provide information beyond directional alignment.

Results and interpretation. At the designated longest window, the six nonzero-proxy configurations have cosines 0.9965–0.9993, projection ratios 0.716–0.938, and relative residuals 0.089–0.291 (Table 6). All six proxy norms exceed the observed norms. At fixed H, N, m, reducing γ moves the projection ratio toward one and decreases the residual. Alternative fixed directions have lower cosine estimates (Appendix C.4.4). The analytical geometry thus describes the observed direction more accurately than its amplitude.

Table 6: Post-hoc comparison of the analytical proxy $p = p_{\gamma,H,m}$ and the 124-dimensional finite-tail mean difference $\widehat{\Delta}_K$ at $K = 64$. Cosine $= p^\top \widehat{\Delta}_K / (\|p\| \|\widehat{\Delta}_K\|)$; projection ratio $= p^\top \widehat{\Delta}_K / \|p\|^2$; relative residual $= \|\widehat{\Delta}_K - p\| / \|p\|$. All six nonzero-proxy configurations are shown; the $H = 1$ checks and sampling intervals are in Appendix C.4.4.

H	N	m	γ/γ_b	Cosine	Projection ratio	Relative residual
2	32	8	1	0.997719	0.938327	0.088505
4	8	32	1	0.996500	0.715730	0.290542
4	32	8	1/4	0.997252	0.935781	0.094644
4	32	8	1/2	0.998972	0.867499	0.138225
4	32	8	1	0.998994	0.784711	0.218151
4	128	2	1	0.999254	0.796703	0.205615

The local-step comparison further separates these two aspects of agreement: at the base step size and $N = 32, m = 8$, the observed $H = 4/H = 2$ projected ratio is 2.927, compared with the analytical ratio 3.5. Together, the results give a directional explanation of the finite-tail algorithm difference while identifying quantitative departures in magnitude and local-step scaling. Appendix C.4.4 gives the sampling intervals and window sensitivity.

7 Conclusion and future directions

Client averaging and control correction act on different parts of stationary mean bias. We made this distinction explicit for SCAFFOLD and matched FedAvg: the algorithms share a γ/N term, while their client-independent γ^2 coefficients separate ordinary local-update and persistent-control contributions. Local second moments coupled with third-order curvature explain the additional control effect, and a uniform remainder identifies the coefficients along arbitrary joint small-step, large-client limits. An exact control translation extends the SCAFFOLD result to affine objective shifts.

Controlled experiments examine these quantitative predictions. Real-data model averaging illustrates that finite-tail mean displacement can be an important component of parameter error, and a post-hoc full-vector comparison finds close directional agreement with the derived curvature–noise structure despite magnitude and local-step discrepancies. The analysis and experiments in this study explain an aspect of federated optimization that leading-order bias and variance alone do not distinguish.

Several directions follow from this coefficient-level characterization:

- **Nonlinear client heterogeneity:** Characterize how client-specific curvature and its interaction with local noise modify stationary bias beyond affine objective shifts.
- **Partial participation:** Determine how client subsampling and delayed control updates affect the stationary coefficients.

- **State-dependent noise:** Extend the analysis to gradient noise whose distribution varies with the local model.
- **Growing local-step counts:** Establish regimes in which H varies jointly with N and γ.
- **Bias-aware control design:** Investigate whether alternative control updates can mitigate second-order bias while accounting for drift correction, variance, and computational cost.

Code availability. The companion repository for the implementations, detailed configurations, numerical records, and reproduction instructions is https://github.com/Duck-YP/stationary-bias-study.

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A Common moment estimates and the SCAFFOLD proof

The proof separates the moment estimates shared by both algorithms from the additional contribution of SCAFFOLD controls. We first bound local fluctuations, then obtain the sharp global raw second moment and a stationary mean-balance identity with a uniform remainder. The SCAFFOLD control covariance completes Theorem 4.1; Appendix B uses the same estimates for FedAvg and the matched attribution.

A.1 Notation and stationary inputs

The three scales used to track errors uniformly in N are

$$s := s_{\gamma,N} = \frac{\gamma}{N} + \gamma^2, \qquad r := r_{\gamma,N} = \frac{\gamma^2}{N} + \gamma^3 = \gamma s, \qquad v := v_{\gamma,N} = \frac{\gamma^2}{N^2} + \gamma^3. \tag{38}$$

Constants denoted by K may change between occurrences and depend on fixed H, d and the objective/noise regularity constants, but not on N or γ. All $O(\cdot)$ terms in these proofs have this same uniform meaning. We use Euclidean norms for vectors and tensors, induced norms for linear maps, and $\|\cdot\|_F$ for the Frobenius matrix norm. In particular, $\|P^h\|$ is an operator norm and operators on matrices use the Frobenius-induced norm. We use repeatedly, for $\gamma \le 1$ and $N \ge 2$,

$$v \le r, \qquad \gamma^2\sqrt{s} \le r, \qquad \gamma^2(s + N^{-1}) \le 2r, \qquad \gamma^4 \le \gamma v. \tag{39}$$

For example, $\sqrt{s} \le \gamma + N^{-1}$ follows by squaring both sides. For vectors u, v, set $(T[u,v])_i = \sum_{j,k} T_{ijk}u_jv_k$. In cubic identities, Sym is the unnormalized sum of three placements:

$$[\mathrm{Sym}(u \otimes M)]_{ijk} = u_iM_{jk} + u_jM_{ik} + u_kM_{ij}, \qquad [\mathrm{Sym}(u^{\otimes 2} \otimes v)]_{ijk} = u_iu_jv_k + u_iv_ju_k + v_iu_ju_k,$$

with M symmetric. This convention extends linearly under expectation.

We use stationary moment bounds from Mangold et al. [9, Corollary B.4 and Lemma B.5] and Mangold et al. [10, Proposition 3 and Lemmas 5–6]. These results apply here because the noise laws are fixed across rounds, $F_\varepsilon(y) = f(y) + \langle \varepsilon, y\rangle$ has the same Hessian as f, and bounded additive noise has the required moments without state-dependent growth. Identical client objectives remove the heterogeneity term. A common sufficient step-size threshold is

$$0 < \gamma \le \gamma_{\mathrm{imp}} := \min\left\{1, \frac{1}{48L}, \frac{1}{24H(L+\mu)}\right\}. \tag{40}$$

It also implies the FedAvg higher-moment restrictions $\gamma \le 1/(9L)$ and $\gamma\mu H \le 1$. The additional SCAFFOLD variance-growth restrictions are vacuous because their parameter β is zero here. We decrease $\gamma_0 \le \gamma_{\mathrm{imp}}$, independently of N, for subsequent contraction and absorption inequalities.

For each algorithm under its own stationary law, the imported estimates give

$$\mathbb{E}\|x\|^{2p} + \frac{1}{N}\sum_c \mathbb{E}\|\theta_{c,h}\|^{2p} = O(\gamma^p), \qquad p = 1,2,3, \quad 0 \le h \le H. \tag{41}$$

For SCAFFOLD, additionally, we have

$$\sup_c \mathbb{E}\|\xi_c\|^6 \le K, \qquad \sum_c \xi_c = 0 \quad \text{pathwise.} \tag{42}$$

In the common lemmas below, write $\xi_c = 0$ for FedAvg. This notation unifies only the within-round local recursions; it does not impose the SCAFFOLD control-refresh rule on FedAvg. In either case, controls are fixed during a round, and fresh noise is independent of the complete round-start state.

We set $\delta_{c,h} = \theta_{c,h} - x$, $\bar{\delta}_h = N^{-1}\sum_c \delta_{c,h}$, $\bar{\varepsilon}_j = N^{-1}\sum_c \varepsilon_{c,j}$, and

$$\begin{aligned} q(y) &= \nabla f(y) - Ay, \qquad P = I - \gamma A, \qquad b = \mathbb{E}[x], \\ m_h &= \frac{1}{N}\sum_c \mathbb{E}[\theta_{c,h}], \qquad \nu_h = \frac{1}{N}\sum_c \mathbb{E}[q(\theta_{c,h})]. \end{aligned} \tag{43}$$

Also, we write

$$M_x = \mathbb{E}[xx^\top], \qquad M_h = \frac{1}{N}\sum_c \mathbb{E}[\theta_{c,h}\theta_{c,h}^\top], \qquad U_h = M_h - M_x, \qquad Q_\xi = \frac{1}{N}\sum_c \mathbb{E}[\xi_c \xi_c^\top]. \tag{44}$$

These are raw second moments, evaluated under each algorithm's own stationary law. In particular, U_h measures the additional second moment accumulated during the h local steps.

A.2 Common coarse and local second-moment estimates

Lemma A.1 (Coarse scales and local second moments)**.** *For either algorithm and* $0 \le h \le H$*,*

$$\frac{1}{N}\sum_c \mathbb{E}\|\delta_{c,h}\|^{2p} = O(\gamma^{2p}), \quad p = 1, 2, 3, \tag{45}$$

$$\mathbb{E}\|x\|^2 + \|b\| = O(s), \qquad m_h - b = O(r), \tag{46}$$

$$\mathbb{E}\|\bar{\delta}_h\|^2 = O(\gamma^2(s + N^{-1})), \tag{47}$$

$$U_h = \gamma^2(hC + h^2 Q_\xi) + O(r). \tag{48}$$

Proof. Unrolling the fixed-H local recursion and using Equations (41) and (42) gives Equation (45) (the control bound is automatic for FedAvg). At the server, the client control corrections cancel by the zero-sum condition. The resulting round increment and stationary balance are

$$x^+ - x = -\gamma \sum_{h=0}^{H-1} \frac{1}{N}\sum_c \nabla f(\theta_{c,h}) - \gamma \sum_{j=1}^{H} \bar{\varepsilon}_j, \tag{49}$$

$$0 = \sum_{h=0}^{H-1} \frac{1}{N}\sum_c \mathbb{E}[\nabla f(\theta_{c,h})]. \tag{50}$$

Coarse global moment and mean. Write $x^+ = \Phi(x) + \zeta + E_{\text{traj}}$, where $\Phi(x) = x - \gamma H \nabla f(x)$, $\zeta = -\gamma \sum_{j=1}^{H} \bar{\varepsilon}_j$, and $E_{\text{traj}} = -\gamma \sum_h N^{-1} \sum_c \{\nabla f(\theta_{c,h}) - \nabla f(x)\}$. Strong convexity gives $\|\Phi(x)\|^2 \le (1 - c\gamma)\|x\|^2$ for some $c > 0$, while $\mathbb{E}\|\zeta\|^2 = O(\gamma^2/N)$ and $\mathbb{E}\|E_{\text{traj}}\|^2 = O(\gamma^4)$. First, we use $\mathbb{E}\langle \Phi(x), \zeta\rangle = 0$, then apply Young's inequality only to E_{traj}. For fixed sufficiently small $a > 0$, we derive

$$\mathbb{E}\|x\|^2 \le (1 + a\gamma)\{(1 - c\gamma)\mathbb{E}\|x\|^2 + K\gamma^2/N\} + K_a \gamma^{-1}\mathbb{E}\|E_{\text{traj}}\|^2.$$

Absorption gives $\mathbb{E}\|x\|^2 = O(s)$ and hence $N^{-1}\sum_c \mathbb{E}\|\theta_{c,h}\|^2 = O(s)$. Since $\|q(y)\| \le K\|y\|^2$, we have $\nu_h = O(s)$. The averaged recursion $m_{h+1} = Pm_h - \gamma\nu_h$, with $m_0 = m_H = b$, gives

$$(I - P^H)b = -\gamma \sum_{j=0}^{H-1} P^{H-1-j}\nu_j, \qquad m_h - b = (P^h - I)b - \gamma \sum_{j=0}^{h-1} P^{h-1-j}\nu_j.$$

Using $\|(I - P^H)^{-1}\| = O(\gamma^{-1})$ and $\|P^h - I\| = O(\gamma)$ proves Equation (46). For the client-averaged displacement, the controls again cancel. Its gradient part has second moment $O(\gamma^2 s)$ and the averaged-noise part has second moment $O(\gamma^2/N)$. Their sum gives Equation (47) without assuming the two parts are independent.

Local second moments. To isolate the effect of fresh noise and persistent controls, write $\theta_{c,h} = x + D_{c,h} + R_{c,h}$ with

$$D_{c,h} = -\gamma \left(h\xi_c + \sum_{j=1}^{h} \varepsilon_{c,j} \right), \qquad R_{c,h} = -\gamma \sum_{\ell=0}^{h-1} \nabla f(\theta_{c,\ell}). \tag{51}$$

Zero-sum cancellation and freshness imply

$$\frac{1}{N}\sum_c \mathbb{E}[xD_{c,h}^\top] = 0, \qquad \frac{1}{N}\sum_c \mathbb{E}[D_{c,h}D_{c,h}^\top] = \gamma^2(hC + h^2 Q_\xi). \tag{52}$$

Only the client-averaged first identity is needed; x and individual controls need not be uncorrelated. Furthermore, $N^{-1}\sum_c \mathbb{E}\|D_{c,h}\|^2 = O(\gamma^2)$ and $N^{-1}\sum_c \mathbb{E}\|R_{c,h}\|^2 = O(\gamma^2 s)$. Cauchy–Schwarz bounds the averaged $xR^\top$, $DR^\top$, and $RR^\top$ terms by $O(\gamma s)$, $O(\gamma^2\sqrt{s})$, and $O(\gamma^2 s)$, respectively. All are $O(r)$ by Equation (39), which proves Equation (48). □

A.3 Common higher-moment bootstrap

The signed third moment is needed to control the cubic term in the stationary Taylor expansion. We obtain it after a fourth-moment estimate.

Lemma A.2 (Fourth and signed third moments)**.** *For either algorithm,*

$$\mathbb{E}\|x\|^4 = O(v), \qquad \|\mathbb{E}[x^{\otimes 3}]\| = O(v). \tag{53}$$

Proof. Unrolling the averaged local recursion gives

$$x^+ = P^H x + \eta + \Delta, \qquad \eta = -\gamma \sum_{j=1}^{H} P^{H-j}\bar{\varepsilon}_j, \quad \Delta = -\gamma \sum_{h=0}^{H-1} P^{H-1-h}\bar{q}_h, \quad \bar{q}_h = \frac{1}{N}\sum_c q(\theta_{c,h}). \tag{54}$$

The centered fresh term η is independent of the round-start state. Expansion of sums of independent centered vector blocks gives

$$\mathbb{E}\|\eta\|^2 = O(\gamma^2/N), \qquad \mathbb{E}\|\eta\|^4 = O(\gamma^4/N^2), \qquad \|\mathbb{E}[\eta^{\otimes 3}]\| = O(\gamma^3/N^2). \tag{55}$$

No symmetry of the noise law is used. Global Lipschitzness of ∇f and bounded $\nabla^3 f$ imply $\|q(y)\| \le K\min\{\|y\|, \|y\|^2\}$. Thus $\|q(y)\|^2 \le K\|y\|^4$ and $\|q(y)\|^4 \le K\|y\|^6$; Jensen's inequality and Equation (41) yield

$$\mathbb{E}\|\Delta\|^2 = O(\gamma^4), \qquad \mathbb{E}\|\Delta\|^4 = O(\gamma^7). \tag{56}$$

In particular, an eighth-moment assumption is unnecessary.

Fourth moment. Let $Y = P^H x + \eta$ and $M_4 = \mathbb{E}\|x\|^4$. Independence and centeredness of η give

$$\mathbb{E}\|Y\|^4 \le \|P^H\|^4 M_4 + K\left\{\mathbb{E}\|x\|^2\mathbb{E}\|\eta\|^2 + \|b\|\sqrt{\mathbb{E}\|\eta\|^2\mathbb{E}\|\eta\|^4} + \mathbb{E}\|\eta\|^4\right\} \le \|P^H\|^4 M_4 + K\gamma v.$$

Use $\|y+z\|^4 \le (1+a\gamma)\|y\|^4 + K_a\gamma^{-3}\|z\|^4$ and $\|P^H\|^4 \le 1 - c\gamma$. For fixed sufficiently small $a > 0$, stationarity implies

$$M_4 \le (1+a\gamma)\{(1-c\gamma)M_4 + K\gamma v\} + K_a\gamma^{-3}\mathbb{E}\|\Delta\|^4.$$

The last term is $O(\gamma^4) \subset O(\gamma v)$; absorption proves $M_4 = O(v)$ and also $\mathbb{E}\|Y\|^4 = O(v)$.

Signed third moment. For $T_3 = \mathbb{E}[x^{\otimes 3}]$, the linear part satisfies

$$\mathbb{E}[Y^{\otimes 3}] = (P^H)^{\otimes 3}T_3 + \mathrm{Sym}\{P^H b \otimes \mathbb{E}[\eta\eta^\top]\} + \mathbb{E}[\eta^{\otimes 3}].$$

The two forcing terms are $O(\gamma v)$ by the coarse mean bound and Equation (55). Set $a_h = N^{-1}\sum_c \|\theta_{c,h}\|^2$. By the fourth-moment bound just proved and Equation (45),

$$\mathbb{E}[a_h^2] \le \frac{1}{N}\sum_c \mathbb{E}\|\theta_{c,h}\|^4 \le K(v+\gamma^4) = O(v), \qquad \mathbb{E}[\|Y\|^2 a_h] \le \sqrt{\mathbb{E}\|Y\|^4 \mathbb{E}[a_h^2]} = O(v).$$

Since $\|\Delta\| \le K\gamma\sum_h a_h$, it follows that $\|\mathbb{E}[Y^{\otimes 2}\otimes\Delta]\| = O(\gamma v)$. The other terms in the cubic expansion obey

$$\mathbb{E}[\|Y\|\|\Delta\|^2] \le \sqrt{\mathbb{E}\|Y\|^2\mathbb{E}\|\Delta\|^4} = O(\gamma^4), \qquad \mathbb{E}\|\Delta\|^3 \le (\mathbb{E}\|\Delta\|^4)^{3/4} = O(\gamma^{21/4}),$$

both $O(\gamma v)$. Hence, $(I-(P^H)^{\otimes 3})T_3 = O(\gamma v)$; its inverse has norm $O(\gamma^{-1})$, proving $T_3 = O(v)$. The sharp global second moment can therefore be established using these higher-moment bounds. □

A.4 Common sharp global second moment

We now compare the nonlinear stationary chain with its linear counterpart. The leading global second moment comes from server-averaged noise; the higher-moment bounds control the nonlinear correction.

Lemma A.3 (Sharp stationary raw second moment)**.** *For either algorithm, the nonlinear forcing in the stationary second-moment equation is $O(\gamma r)$, and*

$$M_x = \frac{\gamma}{N} S_C + O(r). \tag{57}$$

Proof. Linear chain. For the linear chain $x_{\text{lin}}^+ = P^H x_{\text{lin}} + \eta$, $\mathbb{E}[\eta\eta^\top] = (\gamma^2/N)\sum_{k=0}^{H-1} P^k C (P^k)^\top$. Its stationary second-moment equation is the H-fold iteration of $M_{\text{lin}} = P M_{\text{lin}} P^\top + \gamma^2 C/N$. Both equations have unique solutions because P is contractive. In an eigenbasis of A, with eigenvalues a_i,

$$(M_{\text{lin}})_{ij} = \frac{\gamma}{N}\frac{C_{ij}}{a_i + a_j - \gamma a_i a_j}, \qquad (S_C)_{ij} = \frac{C_{ij}}{a_i + a_j}. \tag{58}$$

Consequently $M_{\text{lin}} = (\gamma/N)S_C + O(\gamma^2/N)$ without assuming $AC = CA$.

Nonlinear state correction. To control $\mathbb{E}[x\Delta^\top]$, expand $\bar{q}_h = q(x) + Dq(x)\bar{\delta}_h + \bar{\rho}_h$, where $\|Dq(x)\| \le K\|x\|$ and $\|\bar{\rho}_h\| \le KN^{-1}\sum_c \|\delta_{c,h}\|^2$. The expansion $q(x) = \frac{1}{2}T[x,x] + O(\|x\|^3)$ and the preceding lemmas give

$$\begin{aligned}
\|\mathbb{E}[xq(x)^\top]\|_F &\le K\{\|\mathbb{E}[x^{\otimes 3}]\| + \mathbb{E}\|x\|^4\} = O(r),\\
\mathbb{E}[\|x\|^2\|\bar{\delta}_h\|] &\le \sqrt{\mathbb{E}\|x\|^4\mathbb{E}\|\bar{\delta}_h\|^2} = O\big(\sqrt{v\gamma^2(s+N^{-1})}\big) = O(r),\\
\mathbb{E}[\|x\|\|\bar{\rho}_h\|] &= O(\gamma^2\sqrt{s}) = O(r).
\end{aligned}$$

The outer factor γ in Δ therefore yields

$$\|\mathbb{E}[x\Delta^\top]\|_F = O(\gamma r). \tag{59}$$

Fresh-noise/nonlinear correlation. To control $\mathbb{E}[\eta\Delta^\top]$, hold the complete round-start state fixed. Index the NH fresh blocks by $i = (c,j)$ and write $\eta = \sum_i B_i\varepsilon_i$, with $\|B_i\| \le K\gamma/N$. Let $\Delta^{(i,0)}$ replace block ε_i by zero. Only client c changes before aggregation. Lipschitz propagation gives a path change of at most $K\gamma\|\varepsilon_i\|$ and hence

$$\|\Delta - \Delta^{(i,0)}\| \le K\frac{\gamma^2}{N}\|\varepsilon_i\|, \qquad \mathbb{E}[\varepsilon_i(\Delta^{(i,0)})^\top \mid \text{round-start state}] = 0.$$

Each term $\mathbb{E}[B_i\varepsilon_i(\Delta - \Delta^{(i,0)})^\top]$ is $O(\gamma^3/N^2)$; summing the NH terms gives

$$\|\mathbb{E}[\eta\Delta^\top]\|_F = O(\gamma^3/N). \tag{60}$$

Replacing one noise block at a time handles the dependence between η and Δ. Finally, $\|\mathbb{E}[\Delta\Delta^\top]\|_F = O(\gamma^4)$ by Equation (56).

Stationary covariance equation. For $\mathcal{S}_\gamma(X) = X - P^H X (P^H)^\top$, stationarity gives

$$\begin{aligned}\mathcal{S}_\gamma(M_x - M_{\text{lin}}) = P^H\mathbb{E}[x\Delta^\top] + \mathbb{E}[\Delta x^\top](P^H)^\top \\ + \mathbb{E}[\eta\Delta^\top] + \mathbb{E}[\Delta\eta^\top] + \mathbb{E}[\Delta\Delta^\top].\end{aligned} \tag{61}$$

The right-hand side is $O(\gamma r)$, since both γ^3/N and γ^4 are bounded by γr. Moreover, $\|\mathcal{S}_\gamma^{-1}\| \le (1 - \|P^H\|^2)^{-1} = O(\gamma^{-1})$. Thus $M_x - M_{\text{lin}} = O(r)$, proving Equation (57). □

A.5 Common signed Taylor closure

The global raw second moment determines one part of the mean bias. Stationary gradient balance also involves the second moments along the local trajectories. The following lemma combines both through a Taylor expansion.

Lemma A.4 (Local Taylor tail and stationary mean balance)**.** *For either algorithm,*

$$\left\|\frac{1}{N}\sum_c \mathbb{E}[\theta_{c,h}^{\otimes 3}]\right\| = O(r), \qquad \frac{1}{N}\sum_c \mathbb{E}\|\theta_{c,h}\|^4 = O(v), \tag{62}$$

$$b = -\frac{1}{2}A^{-1}T[M_x] - \frac{1}{2H}A^{-1}T\left[\sum_{h=0}^{H-1} U_h\right] + O(r). \tag{63}$$

Proof. Expanding $\theta_{c,h} = x + \delta_{c,h}$ gives

$$\begin{aligned}\frac{1}{N}\sum_c \mathbb{E}[\theta_{c,h}^{\otimes 3}] = \mathbb{E}[x^{\otimes 3}] + \text{Sym}\{\mathbb{E}[x^{\otimes 2}\otimes\bar{\delta}_h]\} \\ + \text{Sym}\left\{\frac{1}{N}\sum_c \mathbb{E}[x\otimes\delta_{c,h}^{\otimes 2}]\right\} + \frac{1}{N}\sum_c \mathbb{E}[\delta_{c,h}^{\otimes 3}].\end{aligned}$$

The four terms have norms bounded respectively by

$$O(v), \qquad O(\sqrt{v\gamma^2(s+N^{-1})}), \qquad O(\gamma^2\sqrt{s}), \qquad O(\gamma^3),$$

using Cauchy–Schwarz and Lemmas A.1 and A.2. All are $O(r)$. Also $N^{-1}\sum_c \mathbb{E}\|\theta_{c,h}\|^4 \le K\{\mathbb{E}\|x\|^4 + N^{-1}\sum_c \mathbb{E}\|\delta_{c,h}\|^4\} = O(v+\gamma^4) = O(v)$. The signed cubic and absolute fourth-moment bounds now control the higher-order terms in the gradient expansion. With $D_4 = \nabla^4 f(0)$, Taylor's theorem gives

$$\nabla f(u) = Au + \frac{1}{2}T[u,u] + \frac{1}{6}D_4[u,u,u] + \rho_5(u), \qquad \|\rho_5(u)\| \le K\|u\|^4. \tag{64}$$

Here $D_4[u,u,u]$ contracts the fourth derivative against three vectors. Equation (62) controls the signed cubic term and absolute fourth-order remainder, giving

$$\left\|\sum_{h=0}^{H-1}\frac{1}{N}\sum_c \mathbb{E}\left[\nabla f(\theta_{c,h}) - A\theta_{c,h} - \frac{1}{2}T[\theta_{c,h},\theta_{c,h}]\right]\right\| = O(r). \tag{65}$$

Substitute into Equation (50), use $\sum_h m_h = Hb + O(r)$ from Lemma A.1, and write $M_h = M_x + U_h$. This gives $0 = HAb + \frac{1}{2}T[\sum_h M_h] + O(r)$ and proves Equation (63). □

A.6 SCAFFOLD control moment and coefficient assembly

The common lemmas reduce the algorithm-specific calculation to the stationary covariance Q_ξ of the persistent controls. Define

$$G_c = \frac{1}{H}\sum_{h=0}^{H-1}\left\{\nabla f(\theta_{c,h}) - \frac{1}{N}\sum_k \nabla f(\theta_{k,h})\right\}, \qquad E_c = \frac{1}{H}\sum_{j=1}^{H}(\varepsilon_{c,j} - \bar{\varepsilon}_j). \tag{66}$$

The control update in Equation (9) gives $\xi_c^+ = -G_c - E_c$ exactly. Independence across clients and local steps yields $\mathbb{E}[E_c E_c^\top] = (1 - 1/N)C/H$. Smoothness and Equation (45) give $N^{-1}\sum_c \mathbb{E}\|G_c\|^2 = O(\gamma^2)$, while $N^{-1}\sum_c \mathbb{E}\|E_c\|^2 = O(1)$. The $GE^\top$ and $EG^\top$ terms are therefore $O(\gamma)$ by Cauchy–Schwarz, not by independence. Control stationarity implies

$$Q_\xi = Q_{\gamma,N,H} = \frac{1}{H}\left(1 - \frac{1}{N}\right)C + O(\gamma). \tag{67}$$

Inserting this covariance into the local second-moment expansion adds the h^2 contribution specific to persistent controls:

$$U_h = \gamma^2\left(h + \frac{h^2}{H}\right)C + O(r), \tag{68}$$

because the omitted pieces are $O(\gamma^2/N + \gamma^3)$. Finally,

$$\sum_{h=0}^{H-1} U_h = \gamma^2\frac{(H-1)(5H-1)}{6}C + O(r). \tag{69}$$

Combining Equations (57), (63), and (69) yields

$$b^{\mathrm{SCAFFOLD}}_{\gamma,N,H} = -\frac{\gamma}{2N}A^{-1}T[S_C] - \gamma^2\frac{(H-1)(5H-1)}{12H}A^{-1}T[C] + O(r).$$

This proves Theorem 4.1, including the stated uniform remainder.

A.7 Proof of the affine-control translation

Proof of Proposition 4.4. The gradient is $\nabla f_c(y) = \nabla f(y) + a_c$. With the translation in Equation (26), the heterogeneous local step becomes

$$\theta_{c,h+1} = \theta_{c,h} - \gamma\{\nabla f(\theta_{c,h}) + \widetilde{\xi}_c + \varepsilon_{c,h+1}\}.$$

The zero-sum constraint is preserved because $\sum_c a_c = 0$, and the refreshed state satisfies

$$\widetilde{\xi}_c^+ = \xi_c^+ + a_c = \widetilde{\xi}_c + \frac{\theta_{c,H} - x^+}{\gamma H}.$$

Thus the translated controls give exactly the homogeneous round map, with the global model x unchanged. Induction yields pathwise equality for corresponding initial states driven by the same noise blocks.

The translation is invertible and carries invariant laws between the two chains. Uniqueness of the homogeneous invariant law establishes equality of their global stationary marginals. Since $N^{-1}\sum_c f_c = f$, both biases are measured from the same minimizer, and Theorem 4.1 gives the expansion and uniform remainder. □

B FedAvg proof and matched attribution

For FedAvg, persistent controls are absent, so the common estimates of Appendix A apply with $\xi_c = 0$. We verify their inputs and then compute the local second moment that determines the FedAvg coefficient. All expectations are under the FedAvg invariant law, with s, r, v defined in Equation (38).

B.1 Application of the common estimates

The stationary moment estimates of Mangold et al. [10, Proposition 3 and Lemmas 5–6] give Equation (41) under the step-size threshold in Appendix A.1. Equation (10) is the common local recursion with $\xi_c = 0$, and the independent fresh noise satisfies the assumptions of the common lemmas. The two round identities used there become

$$x^+ - x = -\gamma \sum_{h=0}^{H-1} \frac{1}{N} \sum_c \nabla f(\theta_{c,h}) - \gamma \sum_{j=1}^{H} \bar{\varepsilon}_j, \tag{70}$$

$$x^+ = P^H x - \gamma \sum_{j=1}^{H} P^{H-j} \bar{\varepsilon}_j - \gamma \sum_{h=0}^{H-1} P^{H-1-h} \bar{q}_h. \tag{71}$$

These identities establish the same stationary balance and nonlinear round decomposition used in Appendix A. Applying Lemmas A.1 and A.2 first gives the required local and higher-moment bounds. Lemma A.3 then identifies the global raw second moment in Equation (57), and Lemma A.4 supplies the stationary mean-balance identity in Equation (63). Only the local second-moment increment remains to be evaluated.

B.2 Local coefficient and completion of the proof

Without persistent controls, $Q_\xi = 0$ and the stochastic part of the local displacement contains only fresh noise. Equations (51) and (52) therefore reduce to

$$D_{c,h} = -\gamma \sum_{j=1}^{h} \varepsilon_{c,j}, \qquad \frac{1}{N} \sum_c \mathbb{E}[x D_{c,h}^\top] = 0, \qquad \frac{1}{N} \sum_c \mathbb{E}[D_{c,h} D_{c,h}^\top] = \gamma^2 h C.$$

The already proved remainder estimate in Lemma A.1 yields

$$U_h = \gamma^2 h C + O(r), \qquad \sum_{h=0}^{H-1} U_h = \gamma^2 \frac{H(H-1)}{2} C + O(r). \tag{72}$$

Substituting this and $M_x = (\gamma/N) S_C + O(r)$ into Equation (63) gives

$$b_{\gamma,N,H}^{\mathrm{FedAvg}} = -\frac{\gamma}{2N} A^{-1} T[S_C] - \gamma^2 \frac{H-1}{4} A^{-1} T[C] + O_{H,d}\left(\frac{\gamma^2}{N} + \gamma^3\right), \tag{73}$$

proving Theorem 4.2. Subtracting the two explicit second-order coefficients gives

$$B_{20}^{\mathrm{SCAFFOLD}}(H) - B_{20}^{\mathrm{FedAvg}}(H) = -\frac{(H-1)(2H-1)}{12H} A^{-1} T[C] = B_{20}^{\mathrm{ctrl}}(H).$$

The common γ/N term cancels in the matched comparison, while the difference of the two uniform remainders remains $O_{H,d}(\gamma^2/N + \gamma^3)$. This completes Corollary 4.3.

Table 7: Controlled experimental settings. Runs are independent simulation units per configuration; the matched studies use algorithm pairs. The smaller-step path supplements $H = 4$ on the primary joint grid. Structural-check settings are given in Table 19.

Comparison	d; H	Client counts and step sizes	Runs
Primary joint grid	2; 2, 4	$N \in \{12, 16, 24, 32, 48\}$, $\gamma = 1/N$. (77)	8
Smaller-step joint grid	2; 4	$N \in \{64, 96, 128, 192\}$, $\gamma = 1/N$. (78)	8
Fixed-step client count	2; 2	$N \in \{8, 16, 32, 64, 128\}$; $\gamma = 1/24$	8
Five-dimensional	5; 2, 4	$N \in \{32, 48, 64, 96, 128\}$; $\gamma = 1/N$	8
Leading coefficient	2; 1, 2, 4, 8	$N = 32$; $\gamma \in \{1/64, 1/128, 1/256, 1/512\}$	12
Cartesian control coefficient	2; 2, 4, 8	$N \in \{32, 64, 128\}$; $\gamma \in \{1/128, 1/256, 1/512\}$	12
Contraction/noise law	2; 4	$N \in \{64, 128\}$; $\gamma \in \{1/512, 1/1024, 1/2048\}$	12

C Numerical protocols, results, and diagnostics

The appendix groups supporting material by purpose: controlled models and inference (Section C.1), coefficient uncertainty and grid sensitivity (Section C.2), estimator and structural checks (Section C.3), and real-data analyses (Section C.4). Parameter settings are collected in tables; supplementary results use the same design-and-interpretation structure as the main experiments.

C.1 Controlled experimental design and inference

C.1.1 Controlled instances and estimation

The primary nonlinear family provides known curvature, noise covariance, and coefficient targets:

$$f(x) = \tfrac{1}{2} x^\top A x + \alpha \psi(u^\top x), \qquad \psi(0) = 0, \quad \psi'(z) = \log\cosh z, \tag{74}$$

$$\varepsilon = L_C z, \qquad L_C L_C^\top = C, \tag{75}$$

The coordinates of z are independent Rademacher variables, with fresh noise blocks across clients, steps, rounds, and simulation units. The two-dimensional instance uses

$$A = \begin{pmatrix} 1.2 & 0.2 \\ 0.2 & 1.0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1.0 & 0.2 \\ 0.2 & 0.7 \end{pmatrix}, \qquad u = \frac{(1,1)^\top}{\sqrt{2}}, \quad \alpha = 0.25. \tag{76}$$

It has $x^\star = 0$, $AC \neq CA$, and $T[M] = \alpha u (u^\top M u)$. The leading coefficient is

$$B_{11} = (-0.0244594282, -0.0305742852)^\top.$$

Table 7 collects the experimental grids.

Mean-balance estimator. Let $P_\gamma = I - \gamma A$, $q(y) = \nabla f(y) - Ay$, and $\bar{q}_h = N^{-1} \sum_c q(\theta_{c,h})$, using pre-update local states. A round contributes

$$\widehat{b}_{\mathrm{bal}} = -\gamma (I - P_\gamma^H)^{-1} \sum_{h=0}^{H-1} P_\gamma^{H-1-h} \bar{q}_h. \tag{79}$$

Table 8: Run schedules. Burn-in and batch length are measured in communication rounds. The real-data regularization parameter λ is given in Table 20.

Study	Heuristic time scale	Burn-in	Batch length	Batches
Joint grid, fixed step, and $d = 5$	$m_{\mathrm{mix}} = \lceil(\mu_{\mathrm{mix}} H\gamma)^{-1}\rceil$; $\mu_{\mathrm{mix}} = 0.85$ $(d = 2)$, 0.8037572789 $(d = 5)$	$100m_{\mathrm{mix}}$	$20m_{\mathrm{mix}}$	100
Cartesian/local step	$\tau_0 = \lceil(0.85H\gamma)^{-1}\rceil$	$50\tau_0$	$10\tau_0$	64
Contraction/noise law	$\tau = \lceil(0.6H\gamma)^{-1}\rceil$	50τ	10τ	64
Real-data training	$\tau_0 = \lceil(\lambda H\gamma)^{-1}\rceil$	$50\tau_0$	$10\tau_0$	64

Averaging this observable over post-burn-in rounds gives $\widehat{b}$. The linear part of the gradient is handled analytically, so the observable focuses on the nonlinear contribution to the stationary mean. The direct global mean supplies a secondary check in Appendix C.3.1.

For $\delta_{\gamma,N} = \gamma + 1/N$, the joint-grid extrapolation is

$$\widehat{\mathcal{C}}_j = \beta_{0,j} + \beta_{1,j}\delta_{\gamma,N} \tag{80}$$

with the analogous fit for $\widehat{D}_j$. The intercepts are compared with the coefficients defined by the joint small-step, large-client limit. Intervals describe sampling uncertainty conditional on the fitted grid; transient effects and deterministic extrapolation error are assessed through the sensitivity checks below.

C.1.2 Run schedules and sampling uncertainty

Table 8 records the simulation schedules. The time scales use curvature-based heuristics. Within-chain batches are used for accumulation and window comparisons; independent runs are the sampling units for uncertainty estimation.

For the joint-grid studies, the precision criterion is

$$h_j \le \max\{0.0025,\, 0.10|B^{\mathrm{target}}_{20,j}(H)|\},$$

where h_j is the Bonferroni-adjusted Student-t half-width. Every reported run met this criterion at the first 100-batch check; the planned schedule allowed increments of 50 up to 400 batches. Run-validity checks use finite states, a coordinatewise stationarity diagnostic at $z = 3$, at least seven valid units out of eight, and a 10^{-10} control-sum tolerance.

Table 9 summarizes the interval constructions. For the joint-grid fits, a resampled unit is a complete run-level grid vector. Matched outputs stay together, preserving their covariance in coefficient differences. The trigonometric study instead resamples run vectors independently within each configuration, reflecting its independent streams across grid cells.

C.1.3 Cartesian-grid and local-step estimation

The independent study comprises 34 controlled configurations. Each run contributes a grid vector, and OLS is applied runwise before averaging fitted coefficients. The designs are $(1, \gamma)$ for leading coefficients and $(1, \gamma, 1/N)$ for matched control coefficients. A longer trajectory at $(H, N, \gamma) = (4, 64, 1/256)$ supplies shifted and extended windows as within-run sensitivity analyses. The primary estimates use the schedule in Table 8.

C.1.4 Trigonometric contraction and noise-law design

The trigonometric objective makes covariance-dependent cancellation explicit. With $\alpha = 1/20$, define

$$\begin{aligned} f(x) &= \tfrac{1}{2}(x_1^2 + 2x_2^2) + \alpha g(x), \\ g(x) &= -4(\sin x_1 - x_1) + \sin(x_1 + x_2) - (x_1 + x_2) + \sin(x_1 - x_2) - (x_1 - x_2). \end{aligned} \tag{81}$$

Table 9: Sampling-uncertainty procedures. Simultaneous families are specified separately for the listed estimands. All resampling preserves matched algorithm outputs and their coordinates.

Study	Interval construction	Coordinate levels / simultaneous family
Joint grid and fixed step, $d = 2$	Pointwise Student-t (7 degrees of freedom); 2,000 seed-block bootstrap draws for intercepts	Coordinate level 0.975; bootstrap quantiles $(0.0125, 0.9875)$
Five-dimensional	Pointwise Student-t (7 degrees of freedom); 5,000 bootstrap draws	Coordinate level 0.99; quantiles $(0.005, 0.995)$
Cartesian/local step	Student-t (11 degrees of freedom); Bonferroni adjustment; 2,000 bootstrap draws for sensitivity	16 leading coordinates; 6 control coordinates; 12 H contrasts; 2 coordinates for $H = 4 - 3.5(H = 2)$
Contraction/noise law	4,000 within-cell bootstrap draws; centered standardized maximum; marginal and Bonferroni-t sensitivity	36 scalars: 24 coefficient coordinates, 6 law contrasts, 6 covariance contrasts
Real-data allocation	Nominal jackknife-t intervals; adjusted contrast intervals	105 cellwise projections; separate 36 allocation/configuration contrasts
Model averaging	19,999 paired-seed bootstrap draws; centered standardized maximum; delete-one-seed Bonferroni-t sensitivity	Separate primary (196), secondary (392), and diagnostic (1,176) families
Directional comparison	19,999 paired-seed resamples and jackknife sensitivity	Separate 95% scalar (1,068) and vector (117) families

Table 10: Covariance-matched scalar noise laws. Both have mean zero and variance one. Symmetric noise is used at $\rho = -1/4, 0, 1/4$; asymmetric noise is also tested at $\rho = 1/4$.

Law	Support	Probabilities	$\mathbb{E}[z_i^3]$	$\mathbb{E}[z_i^4]$
Symmetric	$(-1, 1)$	$(1/2, 1/2)$	0	1
Asymmetric	$(\sqrt{2}, -1/\sqrt{2})$	$(1/3, 2/3)$	$1/\sqrt{2}$	3/2

Its perturbation has Hessian norm at most $8\alpha = 0.4$, so $0.6I \preceq \nabla^2 f(x) \preceq 2.4I$ globally. Derivatives of orders three to five are bounded, $x^\star = 0$, and $A = \mathrm{diag}(1, 2)$. For symmetric M,

$$T[M] = (2\alpha(M_{11} - M_{22}), -4\alpha M_{12})^\top.$$

Thus the covariance family in Section 6.2.7 gives $S_{C_\rho} = \mathrm{diag}((1+\rho)/2, (1-\rho)/4)$ and $T[C_\rho] = (\rho/5, 0)^\top$. At $H = 4$,

$$\begin{aligned} B_{11}(\rho) &= \left(-\frac{1+3\rho}{80}, 0\right)^\top, \quad B_{20}^{\text{FedAvg}}(4;\rho) = \left(-\frac{3\rho}{20}, 0\right)^\top, \\ B_{20}^{\text{SCAFFOLD}}(4;\rho) &= \left(-\frac{19\rho}{80}, 0\right)^\top, \qquad B_{20}^{\text{ctrl}}(4;\rho) = \left(-\frac{7\rho}{80}, 0\right)^\top. \end{aligned} \tag{82}$$

Noise is $\varepsilon = C_\rho^{1/2} z$, with independent standardized coordinates. Table 10 gives the two bounded laws. They share covariance C_ρ and the common bound $\|\varepsilon\| \le \sqrt{5}$. Fresh draws follow these laws directly. The resolved coefficients depend on covariance, while the remainder allows law differences at order $\gamma^2/N + \gamma^3$.

The full-grid fit uses unweighted OLS with intercept and regressors $(\gamma, 1/N)$. Both normalized individual and matched observables are estimated freely. Noise is shared within each algorithm pair; grid cells, noise laws, pilot runs, and production runs have independent streams. The only restricted-grid sensitivity removes the largest step size.

Table 11: Simultaneous 95% seed-block-bootstrap intervals for the joint-grid coefficients in Table 2. S: SCAFFOLD; F: FedAvg; D: matched difference. P and S identify the primary and smaller-step grids.

Study	H	Grid	Coordinate 1 interval	Coordinate 2 interval
S	2	P	$[-0.0482374, -0.0471965]$	$[-0.0602886, -0.0589883]$
S	4	P	$[-0.1508906, -0.1497518]$	$[-0.1882351, -0.1868165]$
S	4	S	$[-0.1522435, -0.1507781]$	$[-0.1902854, -0.1884530]$
F	2	P	$[-0.0326464908, -0.0313780188]$	$[-0.0408082738, -0.0392220804]$
F	4	P	$[-0.0965774402, -0.0953321996]$	$[-0.1205630991, -0.1190074303]$
F	4	S	$[-0.0964743358, -0.0953790293]$	$[-0.1205859653, -0.1192168647]$
D	2	P	$[-0.0160963688, -0.0159914783]$	$[-0.0201109827, -0.0199811131]$
D	4	P	$[-0.0544427680, -0.0541825657]$	$[-0.0678381597, -0.0675216076]$
D	4	S	$[-0.0559502332, -0.0559007437]$	$[-0.0699256032, -0.0698639209]$

For null and law-difference comparisons, the prespecified resolution margin is 20% of the nonzero first-coordinate target magnitude:

	SCAFFOLD	FedAvg	Matched difference
Margin	0.011875	0.0075	0.004375

Resolution at a margin requires the entire interval to lie inside its symmetric band. These fits use two client counts, so the coefficients of the descriptive grid fit and the asymptotic remainder rate have separate interpretations.

The proof-based step-size bound provides a reference for the tested grids. Several joint-grid steps exceed the imported threshold in Equation (40); all trigonometric steps are below $\gamma_{\text{imp}} = 1/288$. The final sufficient γ_0 also incorporates the contraction and absorption conditions of the proofs.

C.2 Coefficient estimates and finite-grid diagnostics

C.2.1 Joint-grid coefficient intervals

Table 11 supplies the simultaneous intervals for the point estimates and targets in Table 2. The separate individual and paired studies have different sampling precision; their $H = 4$ target-inclusion outcomes are reported separately in Sections 6.2.1 and 6.2.2.

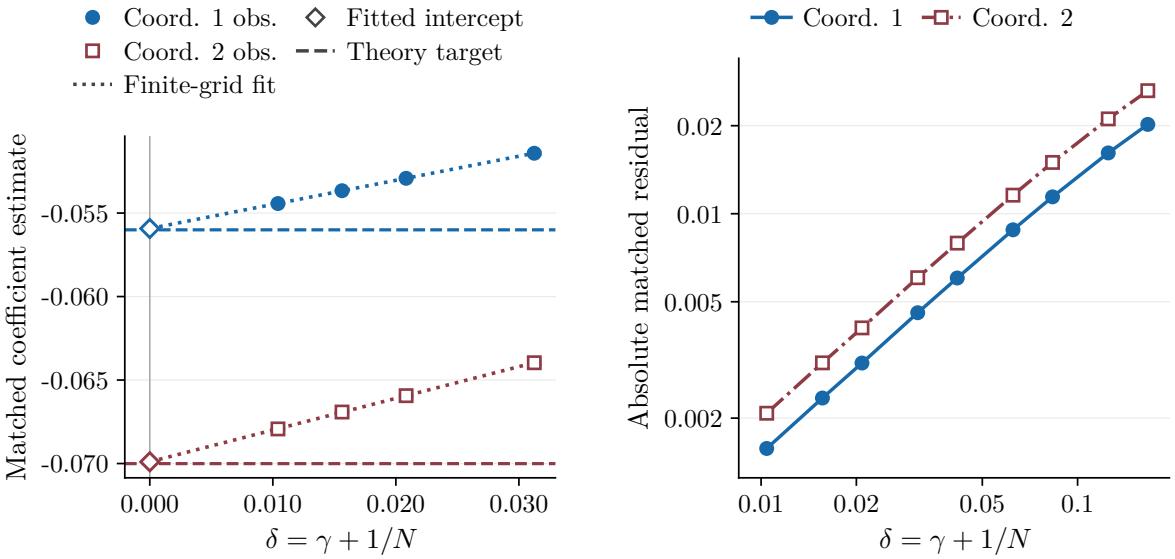
C.2.2 Matched $H = 4$ finite-grid diagnostics

Design and rationale. We examine how the remaining matched error changes with grid resolution. For each point of the combined joint grids, set $e_{\Delta,j} = \widehat{D}_j - B^{\text{ctrl}}_{20,j}(4)$ and compare its magnitude with $\delta = \gamma + 1/N$. A quadratic fit to the same nine points checks sensitivity to the linear extrapolation.

Results and interpretation. Table 12 and Figure 5 show an approximately linear residual trend in δ. The quadratic fit gives intercepts -0.056008582 and -0.070000016, with target errors 3.90×10^{-6} and 5.84×10^{-6}. Thus the extrapolated intercept is sensitive to the finite-grid fit, and the grid-point residuals are compatible with the normalized $O(\gamma + 1/N)$ scale. Since both paths tie γ to $1/N$, the step-size and client-count contributions remain combined. The source of the remaining discrepancy, including finite-run effects, is unresolved; Table 11 retains the primary linear-fit intervals.

Table 12: Matched $H = 4$ grid-point residuals against $\delta = \gamma + 1/N$. The last two columns summarize the observed residual scale across the combined grid.

j	$\lvert e_\Delta \rvert$ at $\delta_{\min}$	$\lvert e_\Delta \rvert$ at $\delta_{\max}$	$\lvert e_\Delta \rvert / \delta$ range	Log–log slope
1	0.0015750	0.0202050	$[0.121, 0.151]$	0.927
2	0.0020785	0.0263460	$[0.158, 0.200]$	0.923



(a) Smaller-step path

(b) Absolute residuals

Figure 5: Matched $H = 4$ finite-grid sensitivity. (a) Smaller-step estimates (circles and squares), finite-grid linear fits (dotted), and analytical targets (dashed). Open diamonds at $\delta = 0$ mark the fitted intercepts. (b) Absolute residuals versus $\delta = \gamma + 1/N$ across all nine points on logarithmic axes; lines connect the observed residuals. The scaled residuals and log–log slopes are given in Table 12. Pointwise bars in (a) use simultaneous Student-t intervals; intercept bars use seed-block-bootstrap intervals.

Table 13: Sampling intervals and offsets for the fixed-step extrapolated intercepts in Figure 2.

j	95% simultaneous CI	Intercept − theory
1	$[-0.0489079, -0.0483504]$	−0.0006102
2	$[-0.0607622, -0.0600656]$	−0.0003902

C.2.3 Fixed-step and five-dimensional results

Design and rationale. These studies separately examine client count and dimension using the objective family in Equation (74). The fixed-step study uses the grid in Table 7. The five-dimensional instance has positive-definite A, C with $\|AC - CA\|_F = 0.0876171216$.

Results and interpretation. Table 13 adds the fixed-step intercept intervals and offsets to Figure 2. Table 14 supplies all five coordinates for each local-step count. These details support the client-count and multidimensional interpretations in Sections 6.2.3 and 6.2.4.

Table 14: Five-dimensional SCAFFOLD coefficient targets, fitted intercepts, and nominal simultaneous 95% bootstrap intervals.

H	j	$\widehat{\beta}_{0,j}$	Theory	95% simultaneous CI
2	1	−0.0140655061	−0.0140249988	[−0.0142709800, −0.0138295249]
2	2	−0.0457104417	−0.0455796216	[−0.0463785328, −0.0449434842]
2	3	0.0322203570	0.0321284384	[0.0316797243, 0.0326913914]
2	4	−0.0340954937	−0.0339978616	[−0.0345938212, −0.0335234201]
2	5	0.0160014114	0.0159557997	[0.0157329196, 0.0162353381]
4	1	−0.0444005615	−0.0444124962	[−0.0446094313, −0.0441636371]
4	2	−0.1441927099	−0.1443354682	[−0.1448724780, −0.1434224100]
4	3	0.1016052163	0.1017400551	[0.1010621370, 0.1020846930]
4	4	−0.1075543334	−0.1076598952	[−0.1080613670, −0.1069797680]
4	5	0.0504598160	0.0505266990	[0.0501901098, 0.0506979332]

Table 15: Independent leading-coefficient fits with nominal 95% intervals. S: SCAFFOLD; F: FedAvg. The targets are $B_{11,1} = -0.02445943$ and $B_{11,2} = -0.03057429$ for every H; $H = 1$ checks the boundary recursion.

H	Alg.	Estimate$_1$	Interval$_1$	Estimate$_2$	Interval$_2$
1	S	−0.024531	[−0.024960, −0.024102]	−0.030664	[−0.031200, −0.030127]
1	F	−0.024531	[−0.024960, −0.024102]	−0.030664	[−0.031200, −0.030127]
2	S	−0.024432	[−0.025068, −0.023795]	−0.030554	[−0.031349, −0.029758]
2	F	−0.024413	[−0.025051, −0.023774]	−0.030525	[−0.031323, −0.029728]
4	S	−0.025169	[−0.025953, −0.024385]	−0.031537	[−0.032517, −0.030557]
4	F	−0.024799	[−0.025574, −0.024024]	−0.031044	[−0.032013, −0.030075]
8	S	−0.028252	[−0.029046, −0.027459]	−0.035610	[−0.036602, −0.034618]
8	F	−0.025913	[−0.026712, −0.025114]	−0.032560	[−0.033559, −0.031562]

C.2.4 Leading-coefficient and restricted-grid sensitivity

Design and rationale. The independent fits estimate the common leading coefficient directly. Restricted-grid comparisons omit the largest step size or, for the matched control fits, the smallest client count. This assesses sensitivity to points farther from the joint asymptotic limit.

Results and interpretation. Table 15 gives the individual leading-intercept estimates, complementing the matched results in Table 3. The $H = 8$ offsets persist as a sensitivity issue even when restricted fits move toward the targets. Primary estimates remain the basis of the comparisons in Section 6.2.5.

C.2.5 Contraction and noise-law resolution

Design and rationale. The supplementary comparisons assess how the matched result relates to individual coefficients, the second coordinate, and restricted grids. All use the model and resolution margins in Appendix C.1.4. The signed-covariance contrast subtracts the $\rho = -1/4$ symmetric condition from $\rho = 1/4$.

Results and interpretation. Table 16 supplements the matched first-coordinate results in Table 4. The paired signed-covariance estimates and simultaneous intervals are

$$\begin{aligned} j = 1: &\quad -0.04372973, \qquad [-0.04378778, -0.04367168], \\ j = 2: &\quad -0.00002676, \qquad [-0.00005541, 0.00000188], \end{aligned}$$

against target $(-0.04375, 0)^\top$. Individual first-coordinate null and noise-law intervals are wider than their resolution bands, so the precision gain is concentrated in the paired increment.

Table 16: Additional contraction and noise-law results: individual coefficients and contrasts, and the paired second coordinate. Estimates and interval endpoints are multiplied by 10^3. S: SCAFFOLD; F: FedAvg; D: matched difference. Targets follow Equation (82). The law contrast is symmetric minus asymmetric at $\rho = 1/4$; the covariance contrast is $\rho = 1/4$ minus $-1/4$ with symmetric noise. Intervals use the primary family in Table 4.

Condition / contrast	Alg.	Estimate_1	Interval_1	Estimate_2	Interval_2
$-1/4$, sym.	S	56.0908	[42.4573, 69.7243]	2.6156	[−4.5559, 9.7871]
$-1/4$, sym.	F	34.2122	[20.5757, 47.8487]	2.6118	[−4.5609, 9.7844]
0, sym.	S	4.6861	[−18.0942, 27.4663]	2.4973	[−6.0117, 11.0064]
0, sym.	F	4.7066	[−18.0689, 27.4822]	2.5118	[−5.9994, 11.0230]
$+1/4$, sym.	S	−46.8325	[−67.9207, −25.7443]	2.2185	[−4.0669, 8.5038]
$+1/4$, sym.	F	−24.9813	[−46.0714, −3.8913]	2.2414	[−4.0463, 8.5291]
$+1/4$, asym.	S	−58.1461	[−81.4153, −34.8769]	2.7156	[−4.1149, 9.5460]
$+1/4$, asym.	F	−36.2643	[−59.5301, −12.9986]	2.7160	[−4.1137, 9.5457]
Law contrast	S	11.3136	[−20.1352, 42.7624]	−0.4971	[−9.8863, 8.8921]
Law contrast	F	11.2830	[−20.1627, 42.7287]	−0.4747	[−9.8649, 8.9156]
Covariance contrast	S	−102.9233	[−127.9666, −77.8799]	−0.3971	[−9.8315, 9.0372]
Covariance contrast	F	−59.1935	[−84.2420, −34.1451]	−0.3704	[−9.8073, 9.0665]
Additional paired second-coordinate results (zero target)					
$-1/4$, sym.	D		—	0.0038	[−0.0175, 0.0251]
0, sym.	D		—	−0.0145	[−0.0349, 0.0059]
$+1/4$, sym.	D		—	−0.0229	[−0.0423, −0.0035]
$+1/4$, asym.	D		—	−0.0005	[−0.0213, 0.0204]
Law contrast	D		—	−0.0224	[−0.0511, 0.0062]

Restricted-grid sensitivity. Table 17 records the three restricted-grid comparisons whose simultaneous intervals narrowly exclude their targets. The two individual coefficient estimates lose sign resolution. All three corresponding Bonferroni-t intervals include the targets, showing sensitivity to the interval construction as well as the grid.

For the $\rho = 1/4$ symmetric paired second coordinate, the primary interval excludes zero but is contained in the resolution band. Its Bonferroni-t and restricted-grid intervals include zero. Reflection symmetry of this objective and symmetric-noise recursion fixes the ideal second-coordinate mean at zero from zero initial states. The small empirical offset therefore probes finite-simulation and interval accuracy, while the resolution-band comparison measures the size of that offset.

First/last-half balance comparisons give eight nominal 95% exclusions among 144 scalar diagnostics, all in individual algorithms; paired intervals contain zero. Direct means have

Table 17: First-coordinate sensitivity after removing the largest step in the contraction study. The first two rows use $\rho = 1/4$ and symmetric noise; the last row is the signed-covariance contrast.

Quantity	Target	Estimate	Simultaneous interval
SCAFFOLD coefficient	-0.059375	-0.01675171	$[-0.05907084, 0.02556741]$
FedAvg coefficient	-0.0375	0.00510137	$[-0.03721159, 0.04741432]$
SCAFFOLD covariance contrast	-0.11875	-0.06840058	$[-0.11872928, -0.01807187]$

Table 18: The four direct–balance discrepancies in the primary FedAvg study. The diagnostic uses the unnormalized difference $|\Delta\widehat{b}|$; the normalized statistics are also shown.

H	N	j	$\widehat{C}_{\mathrm{bal}}$	$\widehat{C}_{\mathrm{dir}}$	$\lvert\Delta\widehat{b}\rvert$	$3\,\mathrm{SE}_{\mathrm{comb}}$
2	16	2	-0.0408082	0.924598	0.00377112	0.00331567
2	48	1	-0.0323449	3.070686	0.00134680	0.00115744
4	16	2	-0.1080240	0.841309	0.00370833	0.00334677
4	48	1	-0.0933450	2.615233	0.00117560	0.00110953

substantially larger sampling variation than balance estimates. The diagnostics assess complementary sources of uncertainty: window comparisons assess temporal variation, finite-window identities assess arithmetic and alignment, and restricted fits assess grid sensitivity. Stationary approximation remains unresolved.

C.3 Estimator and structural consistency checks

C.3.1 Direct–balance estimator checks

Design and rationale. The direct global mean provides a cross-check of the primary nonlinear balance observable. The unnormalized diagnostic is

$$|\widehat{b}_{\mathrm{direct},j} - \widehat{b}_{\mathrm{bal},j}| \le 3\,\mathrm{SE}_{\mathrm{comb}}, \qquad \mathrm{SE}_{\mathrm{comb}} = \sqrt{\mathrm{SE}^2_{\mathrm{direct},j} + \mathrm{SE}^2_{\mathrm{bal},j}}.$$

Using the unnormalized means keeps the comparison on the parameter scale; second-order normalization multiplies estimation errors by γ^{-2}.

Results and interpretation. The primary two-dimensional FedAvg study satisfies the diagnostic in 156 of 160 chain-coordinate checks. Table 18 gives the four exceptions; their discrepancy-to-threshold ratios are 1.137, 1.164, 1.108, and 1.060. All checks satisfy the criterion in the two-dimensional SCAFFOLD studies, fixed-step study, five-dimensional SCAFFOLD study, and smaller-step FedAvg study. These counts concern the individual algorithms. In the independent controlled study, four of 68 algorithm–configuration raw-versus-balance intervals exclude zero. The primary estimates throughout use the balance observable.

Reanalysis of the existing $H = 4$ Cartesian-grid runs gives paired raw-minus-balance fitted-intercept differences of -2.39×10^{-7} and -2.92×10^{-7}. The paired intercepts therefore change little under estimator substitution. Finite-window identities agree where endpoints are available. Missing endpoints and individual noise sums limit this check to those windows; the joint-grid $H = 4$ discrepancy remains as reported in Appendix C.2.2.

C.3.2 Orthogonal coordinate equivariance

Design and rationale. A coordinate change should transform the complete recursion and its coefficient vectors consistently. For $y = Qx$ with $QQ^\top = I$, set $A_y = QAQ^\top$, $C_y = QCQ^\top$, and

Table 19: Structural consistency checks. Corresponding initial states and noise realizations are used within each comparison.

Check	Configuration
Orthogonal rotation	$Q = \begin{pmatrix} 0.8660254037844386 & -0.5 \\ 0.5 & 0.8660254037844386 \end{pmatrix}$ (30°); $H \in \{2, 4\}$; $(N, \gamma) \in \{(16, 1/16), (64, 1/64)\}$; 4 paired runs per configuration, 16 paths in total
Affine translation	$N \in \{8, 32\}$; $H \in \{2, 4, 8\}$; $a_c = \pm\rho u$, $\rho \in \{0, 0.25, 0.5, 1\}$; $\gamma = 1/128$; 24 configurations of 2,000 rounds

$S_{C,y} = QS_CQ^\top$. The Lyapunov equation becomes $A_yS_{C,y} + S_{C,y}A_y = C_y$, and

$$T_y[QMQ^\top] = QT[M] \quad \text{for every symmetric } M. \tag{83}$$

The coefficient formulas then give

$$B_{11,y} = QB_{11}, \qquad B^a_{20,y}(H) = QB^a_{20}(H), \quad a \in \{\text{SCAFFOLD}, \text{FedAvg}, \text{ctrl}\}. \tag{84}$$

The simulation applies the same transformation to every noise draw. Table 19 lists the rotation and affine comparison settings.

Results and interpretation. The full-path residuals satisfy

$$\max_t \|x^{(y)}_t - Qx^{(x)}_t\|_\infty \le 1.1491 \times 10^{-14}, \qquad \max_{t,c} \|\xi^{(y)}_{t,c} - Q\xi^{(x)}_{t,c}\|_\infty \le 1.5765 \times 10^{-14},$$

with $\max_{t,h} \|\bar{q}^{(y)}_{t,h} - Q\bar{q}^{(x)}_{t,h}\|_\infty \le 5.4210 \times 10^{-17}$ and coefficient residuals below 6×10^{-16}. All are within the 10^{-10} tolerance, confirming coordinate consistency of the SCAFFOLD implementation.

C.3.3 Affine translation and update identities

Design and rationale. The affine comparisons use balanced gradient offsets and the translated initial controls prescribed by Proposition 4.4. With common noise, the transformed states should agree at every round. Zero initialization in both untransformed systems gives different transients because the initial states are no longer translated counterparts. Long-run $H = 1$ checks separately assess arithmetic accumulation.

Results and interpretation. The affine residual reported in Section 6.2.6 reproduces the exact translation to numerical precision. Short-run update-identity checks also pass. Over longer runs, 48 of 132 $H = 1$ arithmetic checks exceed 10^{-10}, with maximum vector discrepancy about 7.54×10^{-10}. The shared-objective subset has 24 exceedances among 60 checks and maximum coordinate discrepancy 4.33×10^{-10}. Two additional paths using a stable endpoint-increment calculation reduce vector discrepancies to about 10^{-12}. This improvement is documented for those two paths; the original full-grid outputs remain the reported results.

C.4 Real-data analyses and diagnostics

C.4.1 Objectives, partitions, and allocation estimands

Design and rationale. The shared-objective comparison isolates allocation effects, while disjoint partitions introduce heterogeneous local objectives. We use the a9a training and test

Table 20: Real-data settings. All training configurations use twelve independent SCAFFOLD–FedAvg pairs. Windows within a run share its trajectory. The two longer-run pilots are analyzed separately.

Component	Specification
Training data	32,512 of 32,561 rows selected by a fixed random permutation to allow equal disjoint partitions
Features and penalty	Each 123-feature row divided by $\max\{1, \|z_i\|\}$, then augmented with an intercept; $y_i \in \{0, 1\}$; all 124 parameters regularized with $\lambda = 0.01$
Client objectives	Shared: all clients use the same data; disjoint IID: random partition; label skew: two label-sorted shards per client. Disjoint regimes use one primary and two alternative partitions
Minibatches and base step	Sampling with replacement; $L_{\text{sample}} = 0.51$; $\gamma_b = 0.1/L_{\text{sample}} = 0.1960784314$
Fixed-budget allocations	$Nm = 256$; $(N, m) \in \{(8, 32), (32, 8), (128, 2)\}$; $H = 4$, $\gamma = \gamma_b$
Additional shared-objective runs	Three step-size/local-step variants (Table 5) and three $H = 1$ checks; nine shared-objective configurations in total
Averaging windows	$K \in \{1, 2, 4, 8, 16, 32, 64\}$ final batches per run; primary configuration $H = 4, N = 128, m = 2, \gamma_b$
Primary averaging horizon	1,280 rounds per batch; 6,400 burn-in rounds; longest window 81,920 rounds, ending at round 88,320; 90,439,680 example-gradient evaluations per algorithm
Longer-run pilots	Two runs; 25,600 burn-in rounds and 128 batches of 1,280 rounds; 189,440 rounds per run

sets from the LIBSVM collection.[1] Table 20 collects preprocessing, training, and averaging settings. Each client minimizes

$$f_c(x) = \frac{1}{|D_c|} \sum_{i \in D_c} \{\log(1 + e^{z_i^\top x}) - y_i z_i^\top x\} + \frac{\lambda}{2}\|x\|^2. \tag{85}$$

Equal client weights preserve the global empirical objective. The shared-objective reference aggregate covariance has trace 0.001060 after rounding; disjoint partitions can change reference covariances and local objectives, including under IID assignment.

The reference Newton solve has gradient norm 1.253×10^{-16} and strong-convexity parameter-error bound 1.253×10^{-14}; a separately initialized solver differs by 8.43×10^{-15}. These floating-point residuals quantify numerical accuracy of the reference used to evaluate A_{ref}, T_{ref}, and Σ_{ref}. The projection direction in Equation (35) is fixed before training.

Squared mean displacement is estimated by $\widehat{U}_K$ in Equation (86) at the complete tail. Raw-checkpoint MSE comes from separately accumulated squared errors; its variance estimate subtracts the corresponding cross-run squared-mean estimate. Negative sampling-corrected values are retained. The nominal jackknife-t intervals and adjusted contrast families are described in Table 9.

Results and interpretation. Table 21 adds raw error and held-out loss results to the main allocation study, and supplies the disjoint-partition results. In the label-skew case at $N = 32, H = 4$, deterministic FedAvg converges to a point about 0.08365 from x_{ref}. The algorithm difference therefore includes a substantial drift-correction effect in addition to stochastic mean displacement.

The pointwise projection half-width target 10^{-4} is unmet in 32 of 35 real-data configurations, including the main shared-objective allocations. The separately estimated allocation contrasts

[1] https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/binary.html

Table 21: Additional finite-tail allocation results at $H = 4$, γ_b, and $Nm = 256$. The upper panel supplements Figure 3 with error and loss measurements; the lower panel reports disjoint partitions. Projection intervals are nominal 95% intervals. Raw-checkpoint MSE is multiplied by 10^3; held-out loss is averaged over the per-run tail-average models. S: SCAFFOLD; F: FedAvg.

Shared objective: additional error and loss measurements				
N	MSE (S)	MSE (F)	Test loss (S)	Test loss (F)
8	2.3511	2.3509	0.424981	0.424983
32	2.3629	2.3618	0.424961	0.424970
128	2.3745	2.3655	0.424889	0.424922

N	Alg.	Projection	Nominal interval	MSE	Test loss
Disjoint IID partition					
8	S	0.000341	$[0.000213, 0.000468]$	2.3458	0.424978
8	F	0.000268	$[0.000140, 0.000396]$	2.3454	0.424981
32	S	0.000880	$[0.000638, 0.001123]$	2.3661	0.424963
32	F	0.000472	$[0.000230, 0.000715]$	2.3643	0.424977
128	S	0.003553	$[0.003393, 0.003714]$	2.3642	0.424884
128	F	0.001994	$[0.001830, 0.002158]$	2.3507	0.424936
Disjoint label-skew partition					
8	S	0.000243	$[0.000050, 0.000436]$	2.2526	0.424984
8	F	−0.039298	$[-0.039491, -0.039104]$	5.6791	0.426589
32	S	0.000352	$[0.000143, 0.000561]$	2.2180	0.424981
32	F	−0.055891	$[-0.056096, -0.055686]$	9.0666	0.427274
128	S	0.001256	$[0.001096, 0.001416]$	2.1956	0.424956
128	F	−0.062027	$[-0.062182, -0.061872]$	10.7951	0.427548

nevertheless resolve their signed differences. Precision of an individual mean and precision of a paired contrast are distinct features of the experiment.

C.4.2 Run-length sensitivity

Design and rationale. First/last-half comparisons assess variation within the recorded tails. They use the same 64-batch trajectories and the 2×69 adjustment across controlled and real-data projected contrasts. In the shared-objective real-data study, the primary observable is the direct mean itself.

Results and interpretation. The adjusted first/last-half contrast intervals contain zero. The endpoint allocation contrasts in the first and last halves are 0.003098 and 0.003129 for SCAFFOLD, and 0.002053 and 0.002092 for FedAvg. The similar values support stability of the allocation comparison within the sampled tails. The stronger temporal-equivalence criterion is assessed separately below.

C.4.3 Finite-tail model averaging

Design and rationale. To quantify the components in Equation (36), set $e_{r,K}^{\mathcal{A}} = Y_{r,K}^{\mathcal{A}} - x_{\mathrm{ref}}$ and $\bar{e}_K = R^{-1} \sum_r e_{r,K}$ with $R = 12$. Suppressing the algorithm superscript,

$$\widehat{U}_K = \frac{\sum_{r \neq s} e_{r,K}^\top e_{s,K}}{R(R-1)}, \qquad \widehat{V}_K = \frac{\sum_r \|e_{r,K} - \bar{e}_K\|^2}{R-1}, \qquad \widehat{M}_K = \frac{1}{R} \sum_r \|e_{r,K}\|^2 = \widehat{U}_K + \widehat{V}_K. \quad (86)$$

Table 22: Additional longest-window model-averaging statistics at $K = 64$, with approximate simultaneous 95% primary-family intervals. Parameter errors and training gaps are multiplied by 10^6; test loss is unscaled. The component shares and total-MSE contrast are reported in Section 6.4.

Quantity	SCAFFOLD	FedAvg
$\widehat{M}_K$	17.267 $[15.383, 19.151]$	10.878 $[9.459, 12.297]$
Train objective gap	0.725 $[0.630, 0.819]$	0.361 $[0.295, 0.426]$
Held-out log loss	0.424889 $[0.424879, 0.424899]$	0.424922 $[0.424912, 0.424932]$
Paired difference	Estimate [simultaneous interval]	
$\Delta\widehat{U}_K$	6.378 $[5.848, 6.908]$	
$\Delta\widehat{V}_K$	0.011 $[-0.036, 0.058]$	
Additional intervals for the loss contrasts reported in the main text:		
Δ train gap	$[0.336, 0.393]$	
Δ held-out log loss	$[-0.00003308, -0.00003231]$	

Cross-products between independent runs estimate the squared mean displacement, while between-run deviations estimate the variance of the time-averaged outputs. The estimators retain negative values of $\widehat{U}_K$. Training and test losses are evaluated on each run's own $Y_{r,K}$ and then averaged across runs.

The primary, secondary, and additional configurations use the schedules in Table 20. The $N = 8, 32$ allocations are secondary; six further configurations examine step size, local steps, and $H = 1$. Bootstrap resampling uses one common run-index vector across algorithms, configurations, and overlapping windows. Separate uncertainty families are listed in Table 9; all condition on the fixed dataset and recorded finite-tail protocol.

Results and interpretation. Table 22 gives absolute errors and losses complementing the main-text error shares. For $b_S = b_F + \delta$, the difference in squared means obeys $\|b_S\|^2 - \|b_F\|^2 = 2b_F^\top\delta + \|\delta\|^2$. The sampling-corrected decomposition is $\Delta\widehat{U} = 2(2.57965 \times 10^{-6}) + 1.21856 \times 10^{-6}$, showing the contribution of both the cross term and the squared algorithmic displacement. The total-MSE contrast corresponds to a 58.7% relative difference. For scale, $f(x_{\text{ref}}) = 0.4758413$ and the held-out contrast has magnitude about 0.00769% of reference test loss.

At $N = 32$ and 8, the longest-window SCAFFOLD/FedAvg mean-displacement shares are 14.97%/7.77% and 1.61%/0.97%, respectively. Their squared-mean intervals give weaker evidence against zero than the primary allocation. Together with the main result, these comparisons show that the error share depends on allocation as well as averaging length.

Temporal diagnostics. The predeclared equivalence test requires both a 10^{-3} vector-norm margin and a 10^{-4} projected-drift margin. None of 144 checks meets both requirements. In the primary SCAFFOLD comparison, the first/last 32-batch drift norm is 0.0013633, with upper bound 0.0054372 and projected interval $[-0.0005438, 0.0007484]$. The longest window inherits the 32-batch check because a separate equal-length 64-batch anchor is unavailable. Thus temporal equivalence remains unresolved.

For SCAFFOLD, $\widehat{U}_K \geq \widehat{V}_K$ at $K = 32, 64$, with a positive lower bound for $U - V$ under both interval procedures only at $K = 64$. The prespecified crossover requirement—two consecutive supported windows together with temporal equivalence—is unmet. FedAvg has no point-estimate crossover on this grid. The protocol's 10% parameter-MSE-share threshold is a researcher-chosen reference. The two longer-run pilots in Table 20 leave these temporal questions unresolved. The established result is the finite-tail error decomposition reported in Section 6.4.

C.4.4 Post-hoc directional comparison

Design and rationale. To contextualize the alignment in Table 6, we compare the analytical direction with the fixed reference-parameter direction and the unpreconditioned vector $-T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}]$. Window, step-size, and local-step comparisons assess how alignment and amplitude change within the existing runs. The zero-vector proxy supplies an amplitude baseline. Sampling procedures are listed in Table 9; scalar and vector families each have their own 95% calibration.

Results and interpretation. At $K = 64$, reference-parameter cosines are 0.8953–0.9087 and unpreconditioned-direction cosines are 0.6590–0.6929. Both are lower than the analytical proxy values in the main table; the zero-vector proxy also has larger residuals in all six configurations. These comparisons make the geometric agreement specific to the Hessian-preconditioned curvature–noise direction among the tested references.

For the fixed-H, N, m sequence, adjacent log slopes of the observed projection are 1.8553 and 1.8907. Across shorter averaging windows, the cosine falls as low as 0.872197. These results describe finite-grid and window sensitivity. Allocation comparisons change N and m together, preserving total sampling budget while changing local covariance.

For the $H = 4/H = 2$ ratio in Section 6.5, the exploratory simultaneous interval is $[2.7415, 3.1125]$, the jackknife interval is $[2.7101, 3.1439]$, and the projection-band transform is $[2.6928, 3.1919]$. All exclude 3.5 and include the ordinary local-update ratio 3. The results resolve a departure from the plug-in local-step scaling; attribution of that departure remains open. The $H = 1$ residuals are approximately 10^{-11}–10^{-10}, consistent with the expected zero-proxy numerical identity.

The 54 early/late paired contrasts assess temporal variation of the algorithm difference. The model-averaging diagnostics in Appendix C.4.3 assess each algorithm separately, so the two comparisons address different temporal quantities. The proxy agreement remains a descriptive geometric result for the recorded finite-tail means.